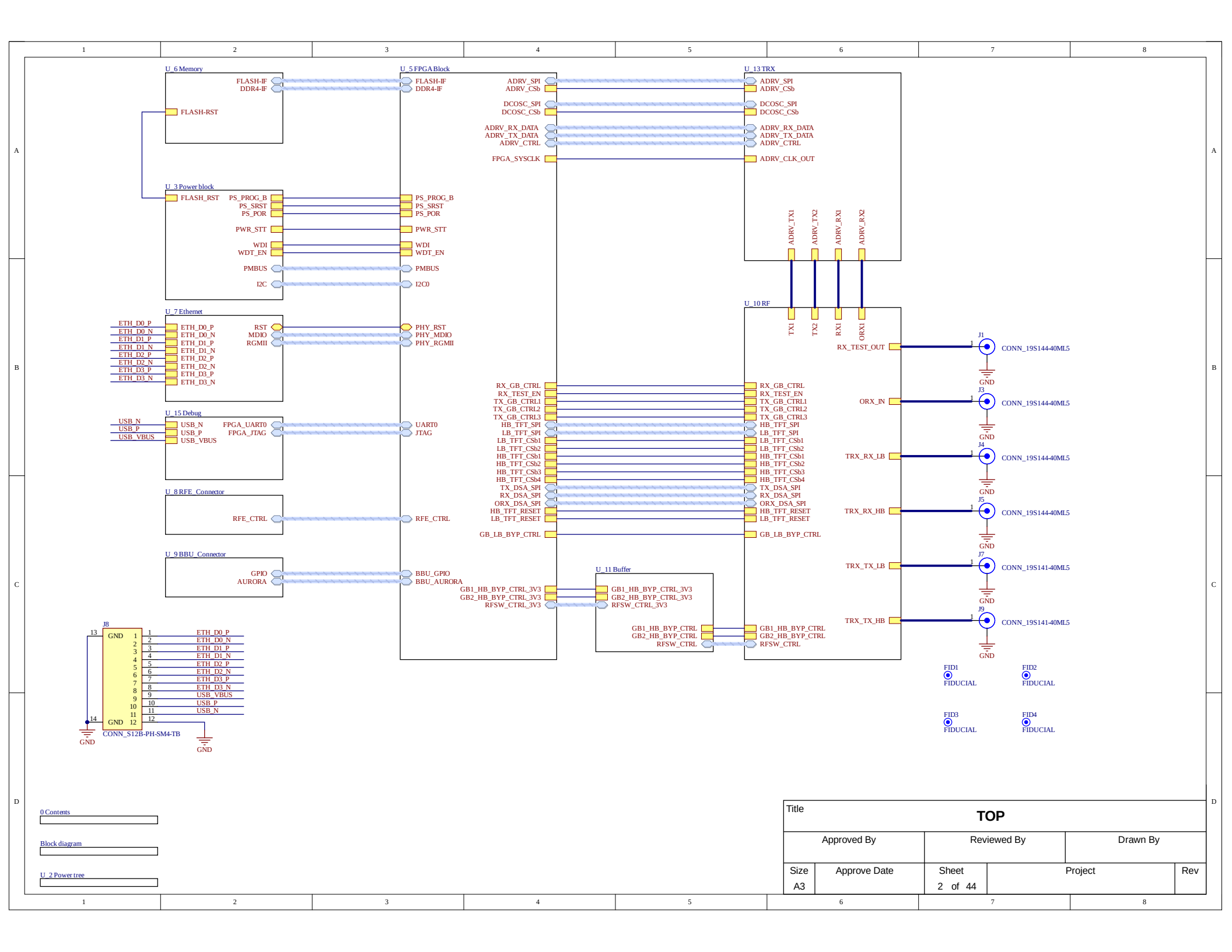
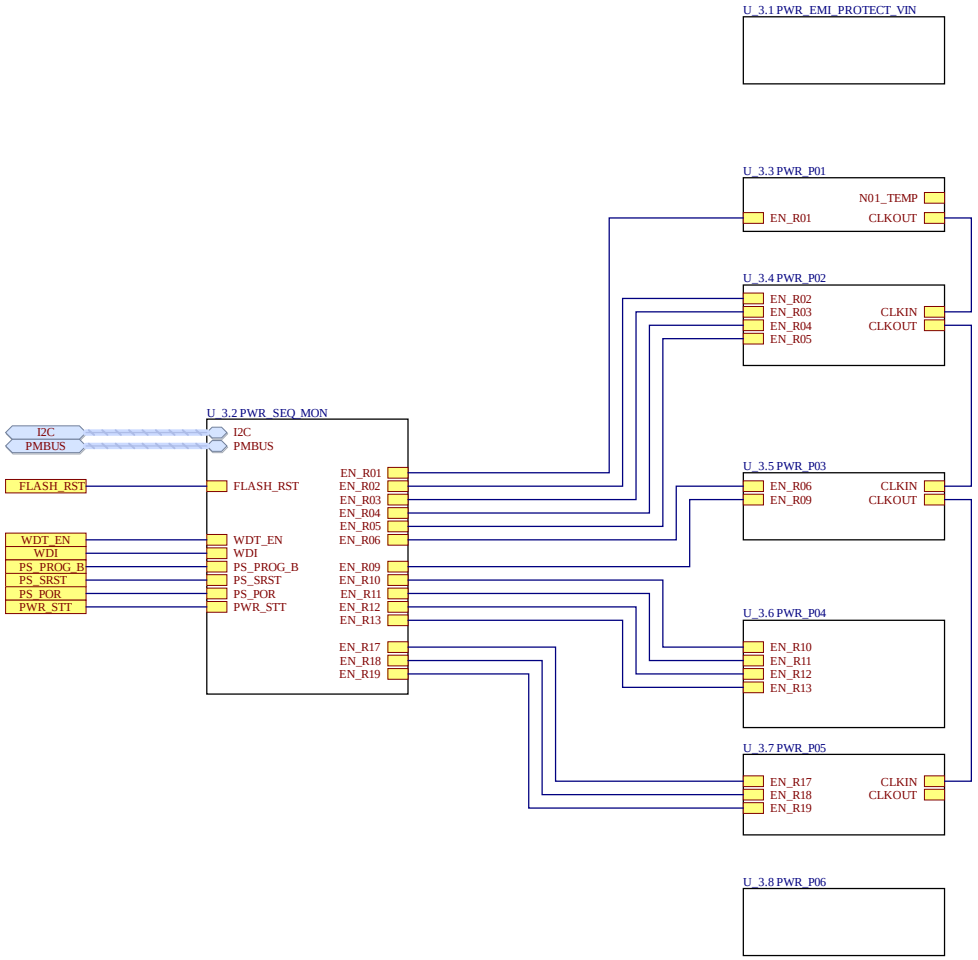
Page	Sheet name	Page	Sheet name
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2	0 Contents	24	5.7 BANK_GTH
3	1 Block diagram	25	5.8 FPGA_PWR1
4	2 Power tree	26	5.9 FPGA_PWR2
5	3 Power block	27	6 Memory
6	3.1 PWR_EMI_PROTECT_VIN	28	6.1 DDR4_P01
7	3.2 PWR_SEQ_MON	29	6.2 DDR4_P02
8	3.3 PWR_P01	30	6.3 eMMC
9	3.4 PWR_P02	31	6.4 Flash_SDcard_EEPROM
10	3.5 PWR_P03	32	7 Ethernet
11	3.6 PWR_P04	33	7.1 ETH_PHY1
12	3.7 PWR_P05	34	7.2 ETH_PHY2
13	3.8 PWR_P06	35	7.3 ETH_TRAN
14	4 Clock	36	13 TRX
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16	5 FPGA Block	38	13.2 ADRV_PWR
17	5.1 BANK_500_501_505	39	13.3 RF_TX_Path
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20	5.3 BANK_504	42	14 Extension connector
21	5.4 BANK_44_47_48_49_50	43	16 RS422
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DDR4-IF
FLASH-RST
U_5 FPGA Block
FLASH-IF
DDR4-IF
U_13 TRX
ADRV_SPI
ADRV_CSb
DCOSC_SPI
DCOSC_CSb
ADRV_RX_DATA
ADRV_TX_DATA
ADRV_CTRL
FPGA_SYSCLK
ADRV_CLK_OUT
U_3 Power block
FLASH_RST
PS_PROG_B
PS_SRST
PS_POR
PWR_STT
WDI
WDT_EN
PMBUS
I2C
I2C0
ADRV_TX1
ADRV_TX2
ADRV_RX1
ADRV_RX2
U_7 Ethernet
ETH_D0_P
ETH_D0_N
ETH_D1_P
ETH_D1_N
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ETH_D2_N
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RST
MDIO
RGMII
PHY_RST
PHY_MDIO
PHY_RGMII
U_10 RF
TX1
TX2
RX1
ORX1
RX_TEST_OUT
J1
CONN_19S144-40ML5
GND
ORX_IN
J3
TRX_RX_LB
J4
TRX_RX_HB
J5
TRX_TX_LB
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CONN_19S141-40ML5
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J9
U_15 Debug
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USB_P
USB_VBUS
FPGA_UART0
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UART0
JTAG
RX_GB_CTRL
RX_TEST_EN
TX_GB_CTRL1
TX_GB_CTRL2
TX_GB_CTRL3
HB_TFT_SPI
LB_TFT_SPI
LB_TFT_CSb1
LB_TFT_CSb2
HB_TFT_CSb1
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HB_TFT_CSb3
HB_TFT_CSb4
TX_DSA_SPI
RX_DSA_SPI
ORX_DSA_SPI
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LB_TFT_RESET
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U_8 RFE_Connector
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GPIO
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BBU_GPIO
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U_11 Buffer
GB1_HB_BYP_CTRL_3V3
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RFSW_CTRL
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GND
CONN_S12B-PH-SM4-TB
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FID2
FID3
FID4
FIDUCIAL
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Title	1 Block diagram		
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Title	2 Power tree		
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R01: +0V85_VCCINT

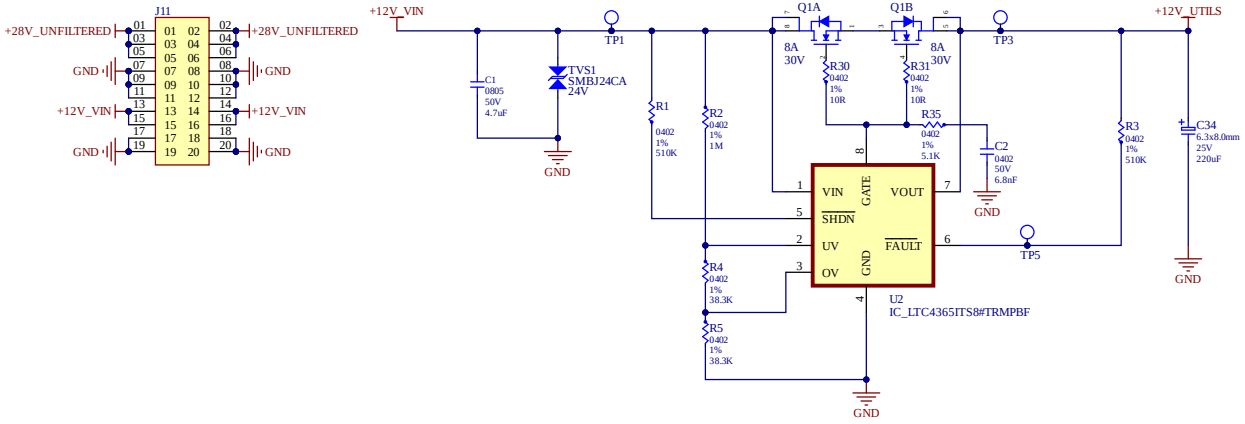
R02: +1V8_VCCAUX
R03: +3V3_VCCO
R04: +0V9_VMGTAVCC
R05: +1V8_VMGTVCCAUX

R06: +1V2_VMGTAVTT
R09: +1V2_DDR4_VDD

R10: +1V2_VCC_PSPLL
R11: +0.85V_VPS_MGTRAVCC
R12: +2V5_DDR4_VPP
R13: +0V6_DDR4_VTT

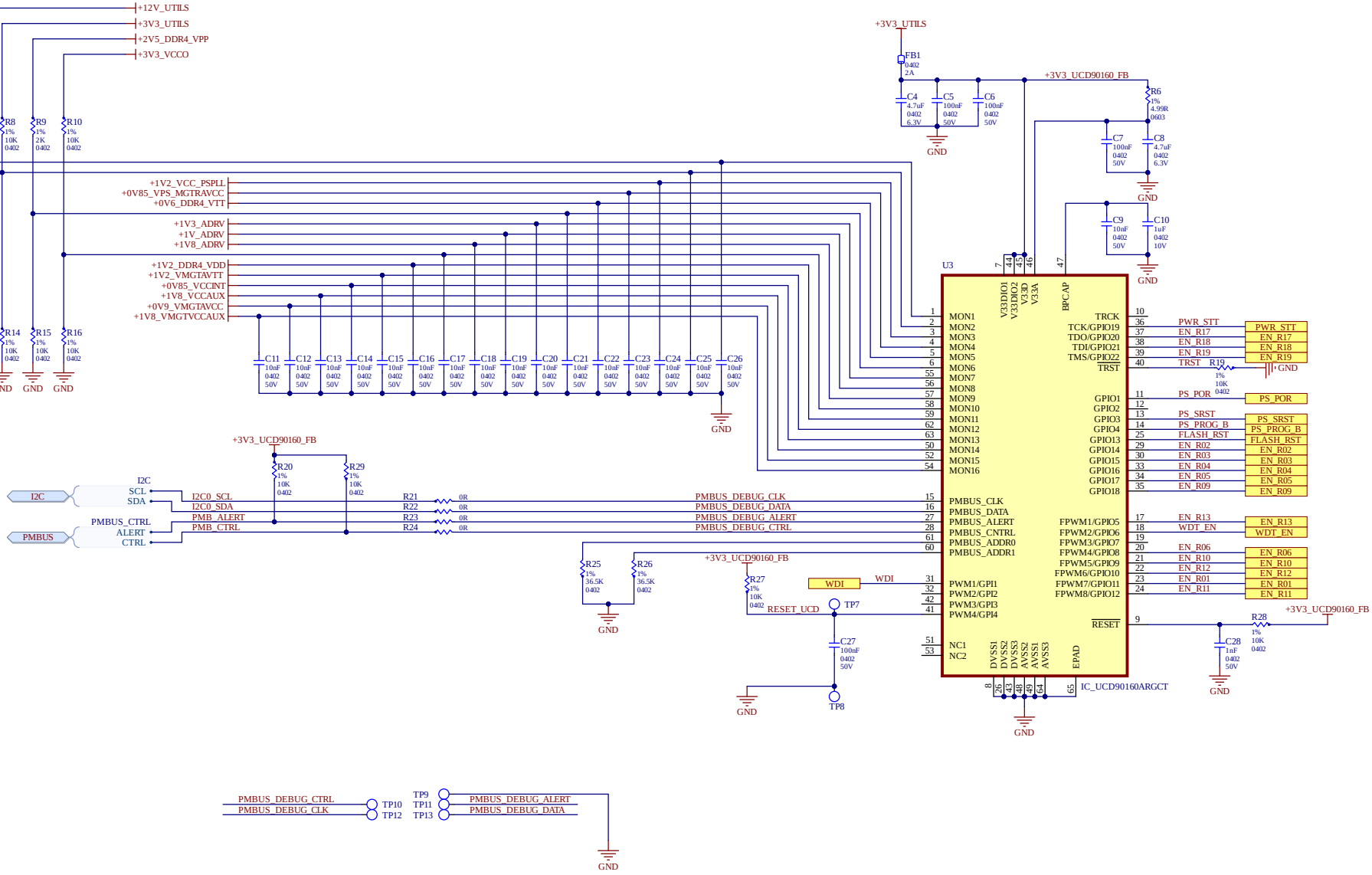
R17: +1V8_ADRV
R18: +1V_VDIG_ADRV
R19: +1V3_ADRV

Title	**3 Power block**			
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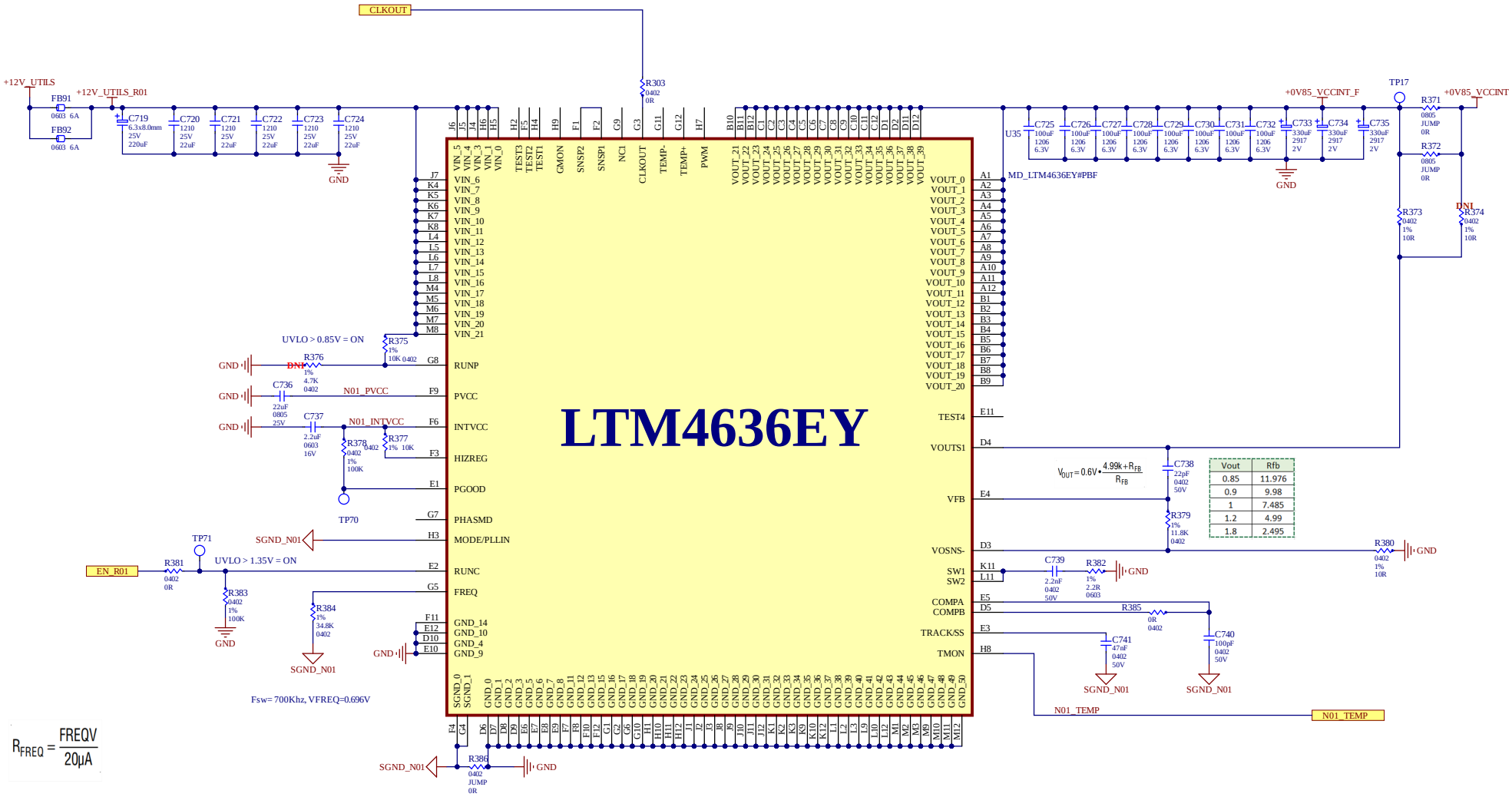
Vtyp	12	V			
Vov	14	V			
Vuv	7	V			
R1	1000	Kohm	Vuv	0.498049	V
R2	38.3	Kohm	Vov	0.498049	V
R3	38.3	Kohm			

Title	3.1 PWR_EMI_PROTECT_VIN			
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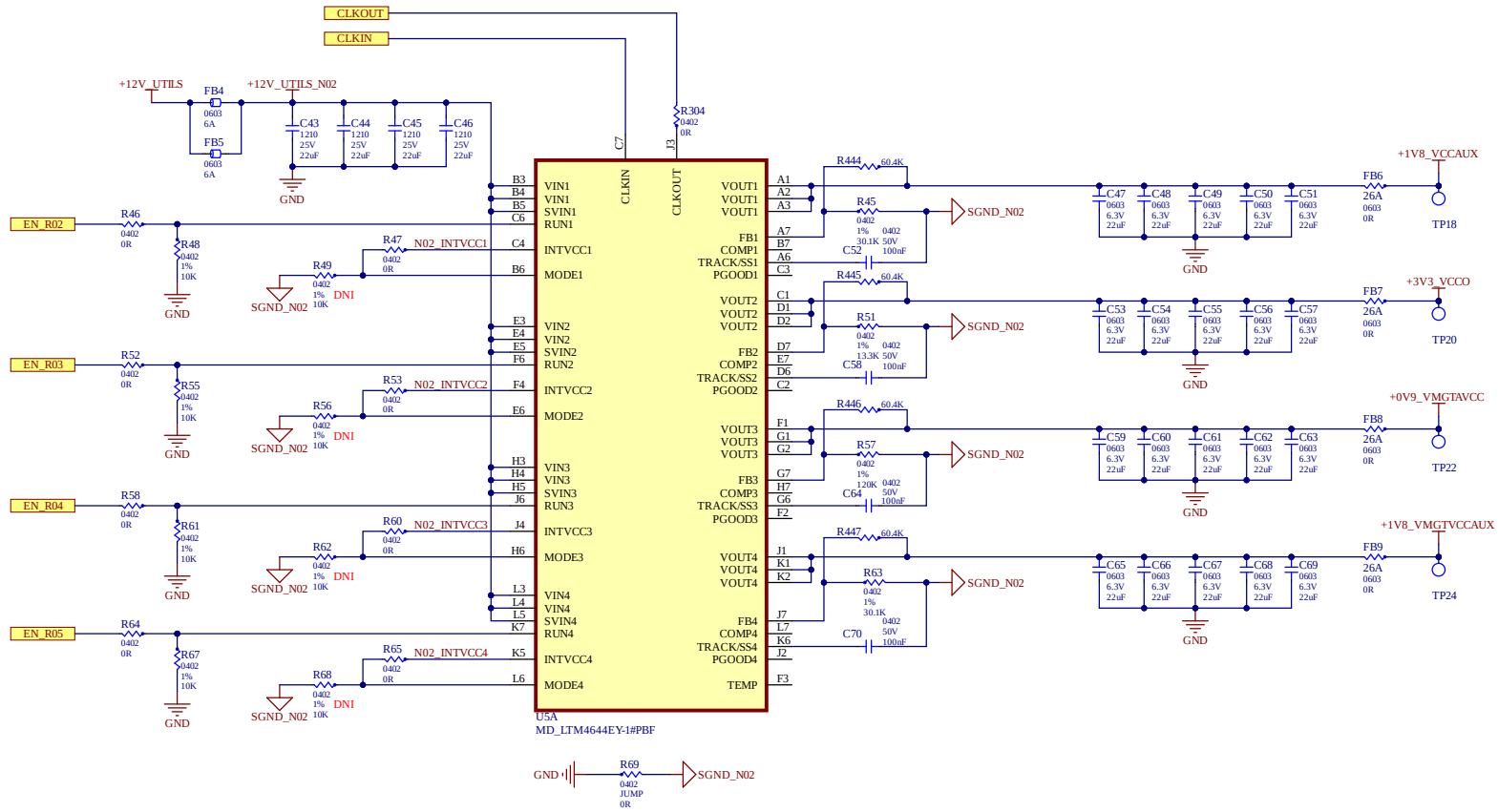
Title	**3.2 PWR_SEQ_MON**		
Approved By		Reviewed By	Drawn By
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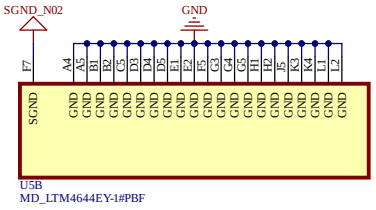
R01: +0V85_VCCINT - 40A



Vout	Rfb
0.85	11.976
0.9	9.98
1	7.485
1.2	4.99
1.8	2.495

Title	3.3 PWR_P01		
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LTM4644	
Rfb (Kohm)	Vout (R)
145	0.85
120	0.90
60.4	1.20
30.1	1.80
13.3	3.32
12	3.62

$$t_{SS} = 0.6 \bullet \frac{C_{SS}}{2.5\mu A}$$

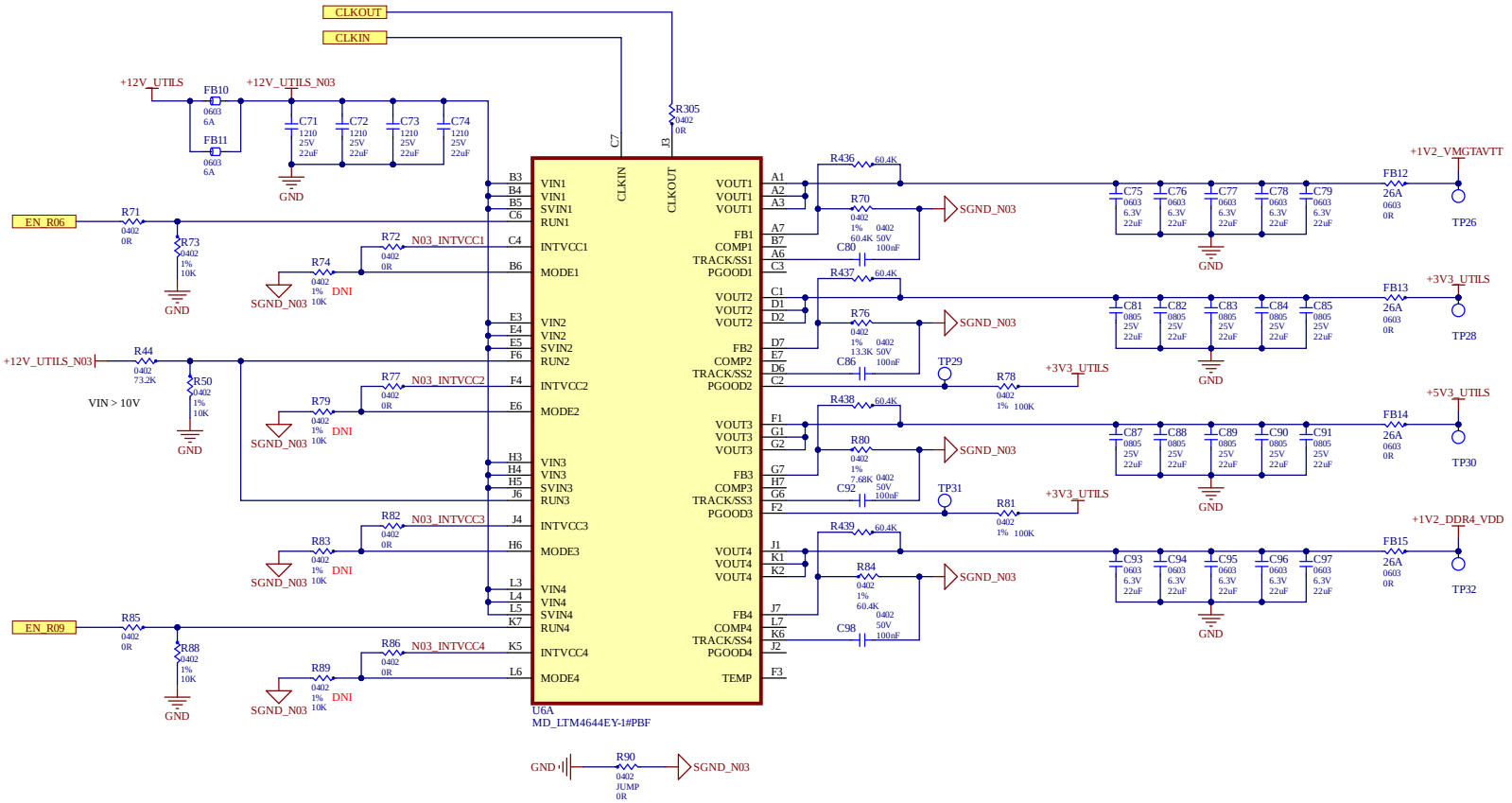
$$R_{FB(BOT)} = \frac{60.4k}{\frac{V_{OUT}}{0.6} - 1}$$

Table 1. V_{FB} Resistor Table vs Various Output Voltages

V_{OUT} (V)	0.6	1.0	1.2	1.5	1.8	2.5	3.3	5.0
$R_{FB(BOT)}$ (k)	Open	90.9	60.4	40.2	30.1	19.1	13.3	8.25

V_{IN}, SV_{IN} (Per Channel) ..... –0.3V to 15V
V_{OUT} (Per Channel) (Note 3) ..... –0.3V to SV_{IN} or 6V
RUN (Per Channel) ..... –0.3V to 15V
$INTV_{CC}$ (Per Channel) ..... –0.3V to 3.6V
PGOOD, MODE, TRACK/SS,
FB (Per Channel) ..... –0.3V to $INTV_{CC}$
CLKOUT (Note 3), CLKIN ..... –0.3V to $INTV_{CC}$
Internal Operating Temperature Range (Note 2, 5)
E and I-Grade ..... –40°C to 125°C
MP-Grade ..... –55°C to 125°C
Storage Temperature Range ..... –65°C to 150°C
Peak Solder Reflow Body Temperature ..... 245°C

Title	3.4 PWR_P02		
Approved By		Reviewed By	Drawn By
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R06: +1V2_VMGTAVTT

R07: +3V3_UTILS

R08: +5V3_UTILS

R09: +1V2_DDR4_VDD

LTM4644	
Rfb (Kohm)	Vout (R)
145	0.85
120	0.90
60.4	1.20
30.1	1.80
13.3	3.32
12	3.62

$$t_{SS} = 0.6 \bullet \frac{C_{SS}}{2.5\mu A}$$

$$R_{FB(BOT)} = \frac{60.4k}{\frac{V_{OUT}}{0.6} - 1}$$

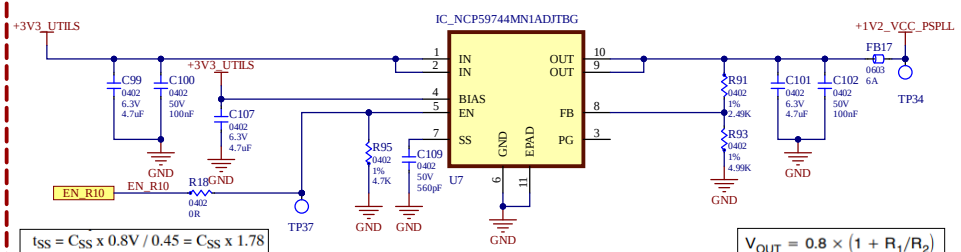
Table 1. V_{FB} Resistor Table vs Various Output Voltages

V_{OUT} (V)	0.6	1.0	1.2	1.5	1.8	2.5	3.3	5.0
$R_{FB(BOT)}$ (k)	Open	90.9	60.4	40.2	30.1	19.1	13.3	8.25

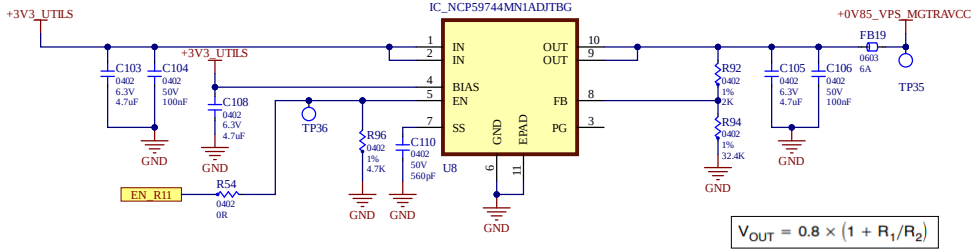
V_{IN}, SV_{IN} (Per Channel)	−0.3V to 15V
V_{OUT} (Per Channel) (Note 3)	−0.3V to SV_{IN} or 6V
RUN (Per Channel)	−0.3V to 15V
$INTV_{CC}$ (Per Channel)	−0.3V to 3.6V
PGOOD, MODE, TRACK/SS, FB (Per Channel)	−0.3V to $INTV_{CC}$
CLKOUT (Note 3), CLKIN	−0.3V to $INTV_{CC}$
Internal Operating Temperature Range (Note 2, 5)	
E and I-Grade	−40°C to 125°C
MP-Grade	−55°C to 125°C
Storage Temperature Range	−65°C to 150°C
Peak Solder Reflow Body Temperature	245°C

Title: 3.5 PWR_P03			
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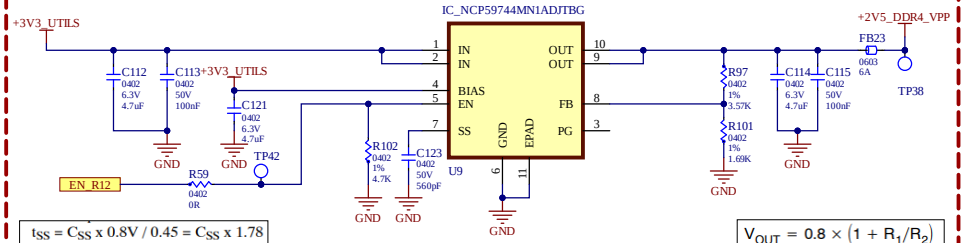
R10: +1V2_VCC_PSPLL - 3A



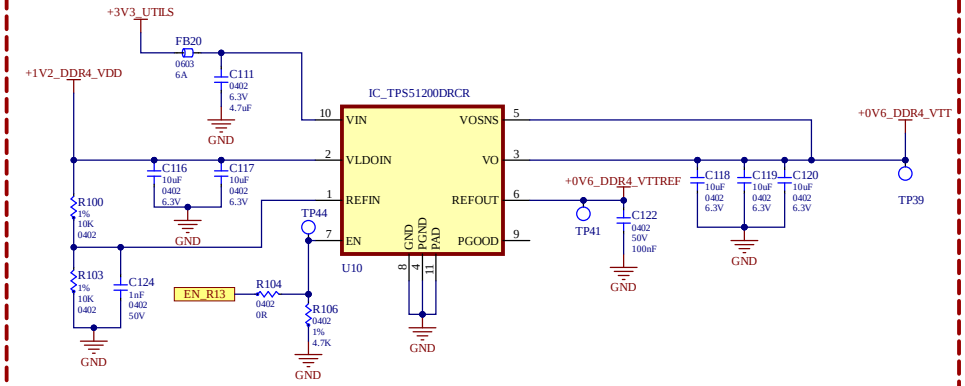
R11: +0.85V_VPS_MGTRAVCC - 3A



R12: +2V5_DDR4_VPP - 3A

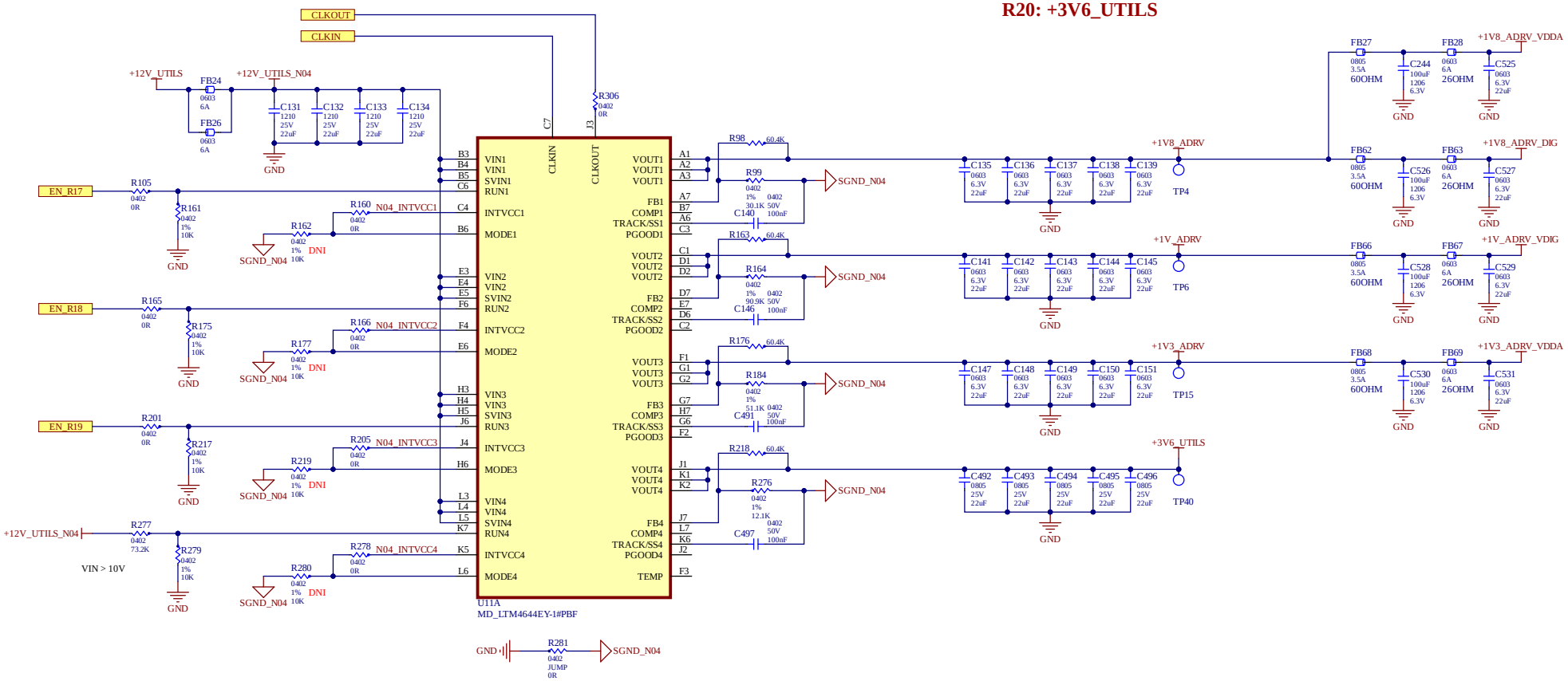


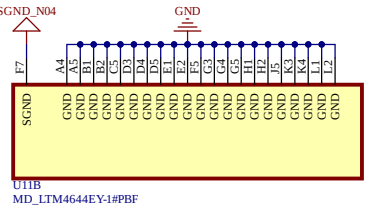
R13: +0V6_DDR4_VTT - 3A
+0V6_DDR4_VTTREF - 1mA



Title	3.6 PWR_P04		
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R17: +1V8_ADRV
R18: +1V_ADRV
R19: +1V3_ADRV
R20: +3V6_UTILS





LTM4644	
Rfb (Kohm)	Vout (R)
145	0.85
120	0.90
60.4	1.20
30.1	1.80
13.3	3.32
12	3.62

$$t_{SS} = 0.6 \bullet \frac{C_{SS}}{2.5\mu A}$$

$$R_{FB(BOT)} = \frac{60.4k}{\frac{V_{OUT}}{0.6} - 1}$$

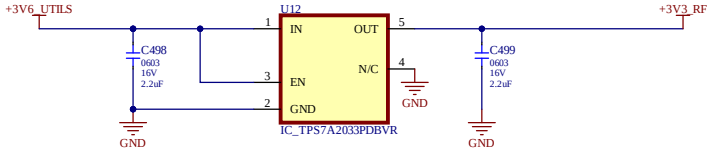
Table 1. V_{FB} Resistor Table vs Various Output Voltages

V_{OUT} (V)	0.6	1.0	1.2	1.5	1.8	2.5	3.3	5.0
$R_{FB(BOT)}$ (k)	Open	90.9	60.4	40.2	30.1	19.1	13.3	8.25

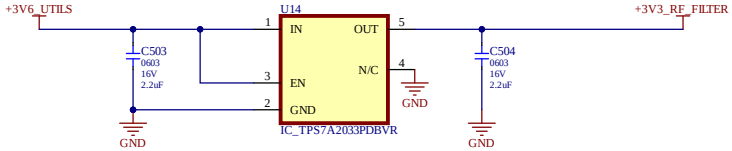
V_{IN}, SV_{IN} (Per Channel) ..... –0.3V to 15V
V_{OUT} (Per Channel) (Note 3) ..... –0.3V to SV_{IN} or 6V
RUN (Per Channel) ..... –0.3V to 15V
$INTV_{CC}$ (Per Channel) ..... –0.3V to 3.6V
PGOOD, MODE, TRACK/SS,
FB (Per Channel) ..... –0.3V to $INTV_{CC}$
CLKOUT (Note 3), CLKIN ..... –0.3V to $INTV_{CC}$
Internal Operating Temperature Range (Note 2, 5)
E and I-Grade ..... –40°C to 125°C
MP-Grade ..... –55°C to 125°C
Storage Temperature Range ..... –65°C to 150°C
Peak Solder Reflow Body Temperature ..... 245°C

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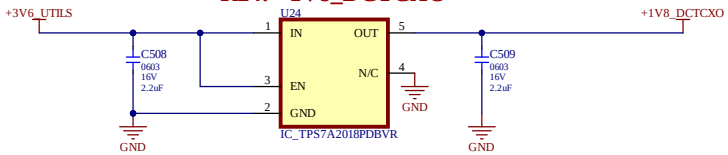
R22: +3V3_RF_SW



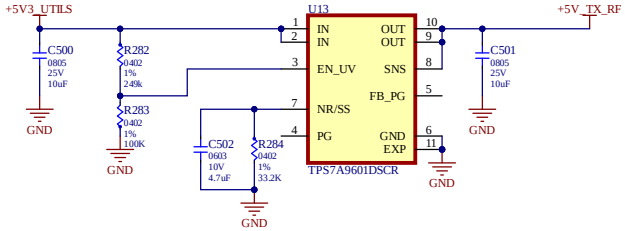
R23: +3V3_RF_FILTER



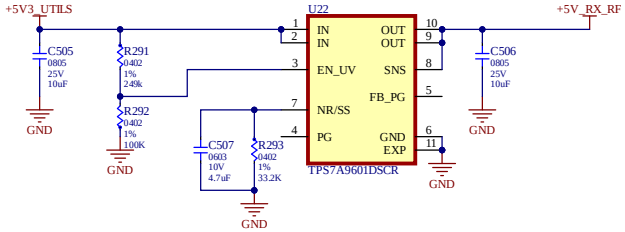
R24: +1V8_DCTCXO



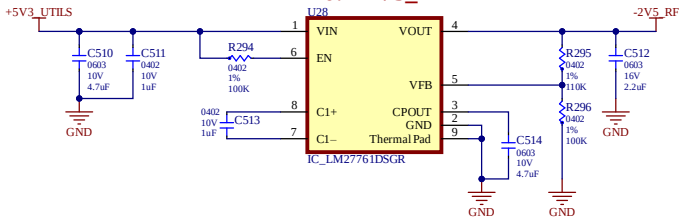
R14: +5V_TX_RF



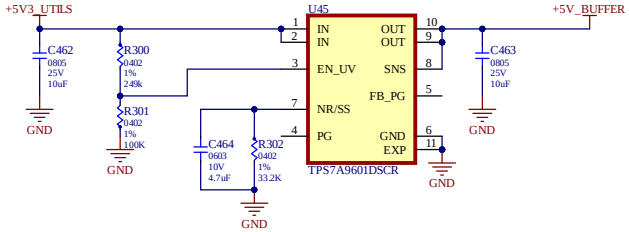
R15: +5V_RX_RF



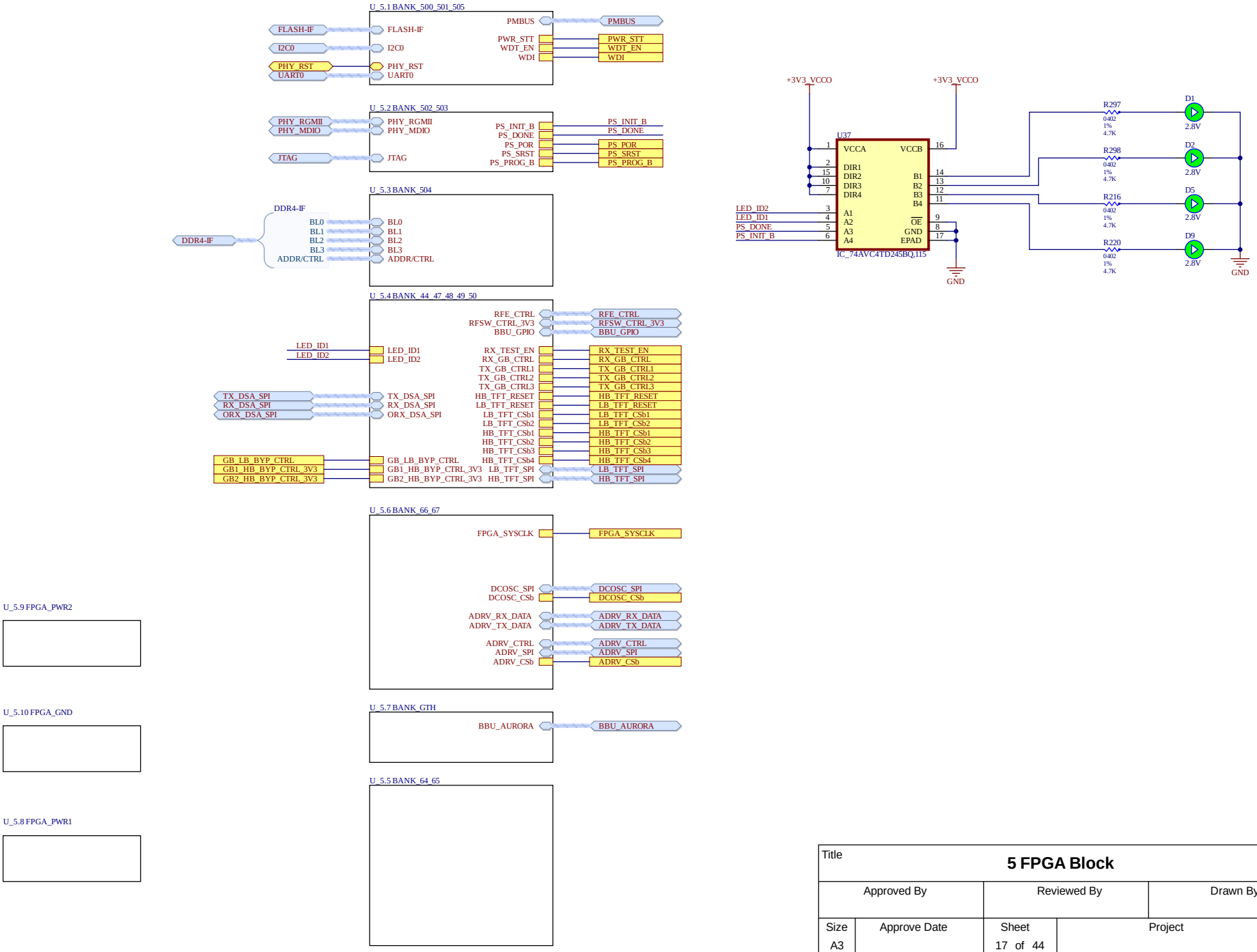
R16: -2V5_RF



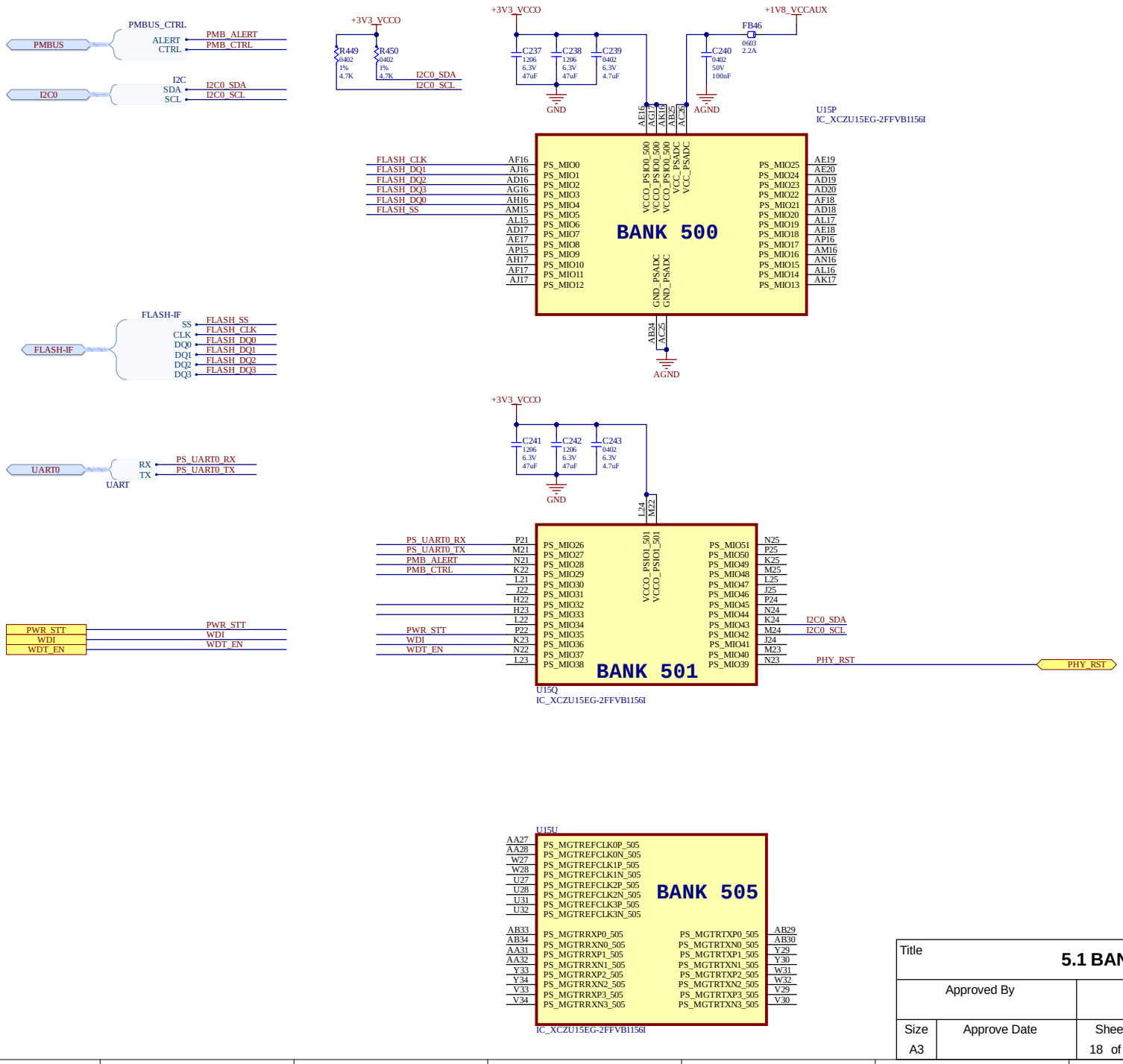
+5V_BUFFER



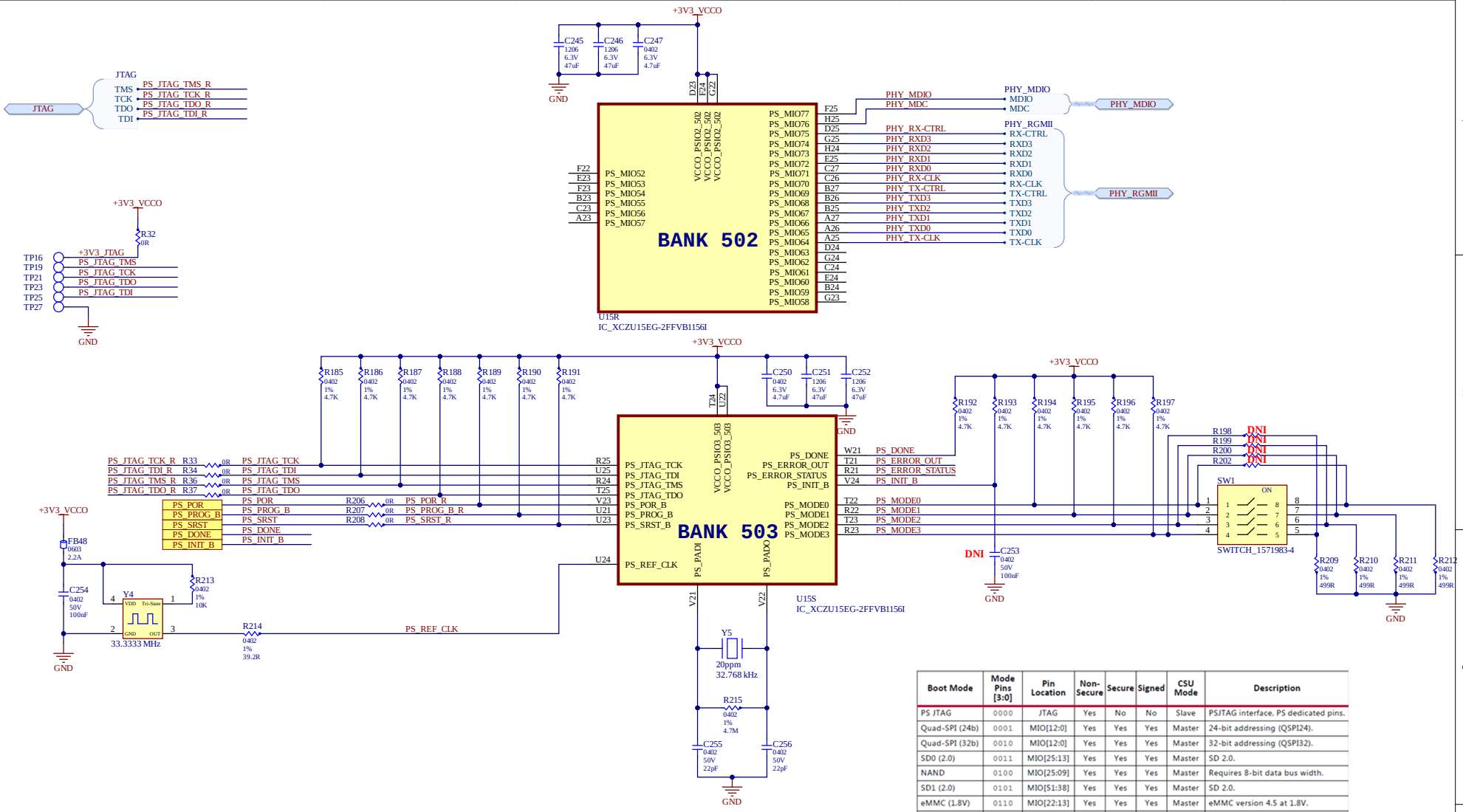
Title	**3.8 PWR_P06**			
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Title	**5 FPGA Block**		
Approved By		Reviewed By	Drawn By
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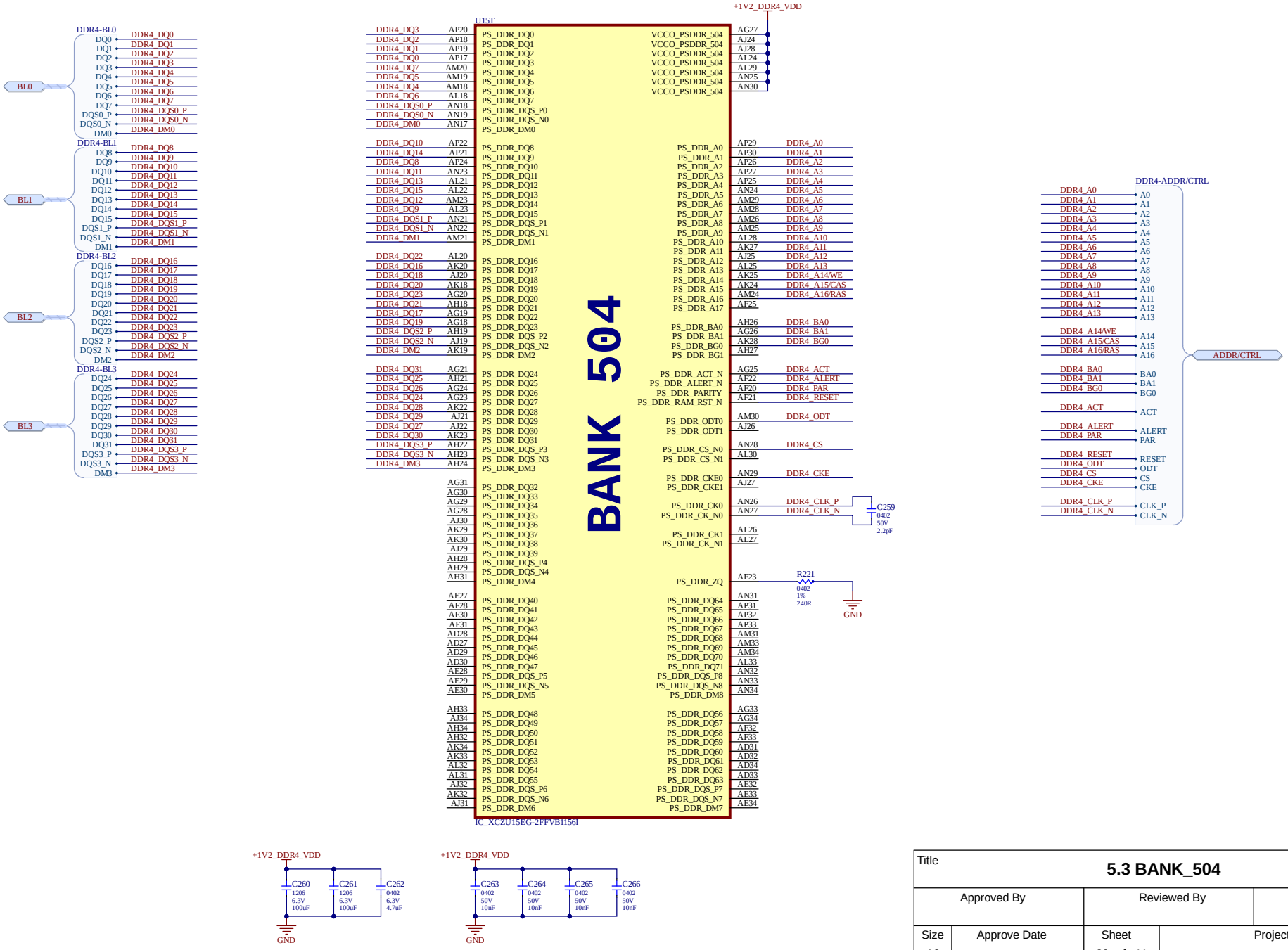


Title	5.1 BANK_500_501_505		
Approved By		Reviewed By	Drawn By
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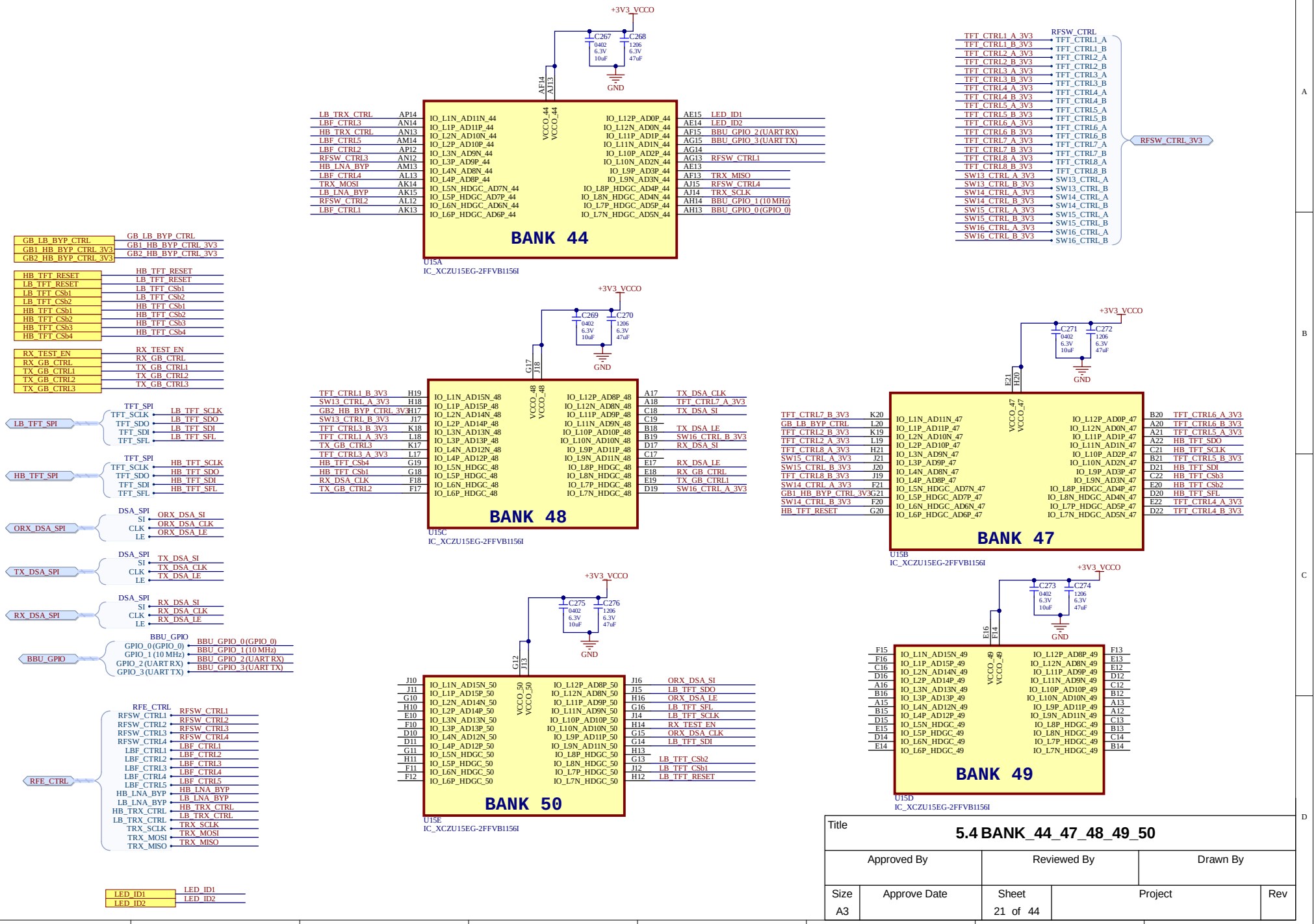


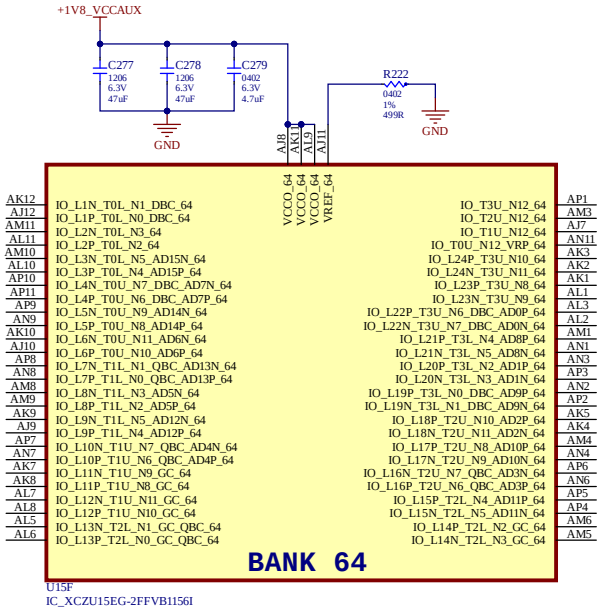
Boot Mode	Mode Pins [3:0]	Pin Location	Non-Secure	Secure	Signed	CSU Mode	Description
PS JTAG	0000	JTAG	Yes	No	No	Slave	PSJTAG interface. PS dedicated pins.
Quad-SPI (24b)	0001	MIO[12:0]	Yes	Yes	Yes	Master	24-bit addressing (QSPI24).
Quad-SPI (32b)	0010	MIO[12:0]	Yes	Yes	Yes	Master	32-bit addressing (QSPI32).
SD0 (2.0)	0011	MIO[25:13]	Yes	Yes	Yes	Master	SD 2.0.
NAND	0100	MIO[25:09]	Yes	Yes	Yes	Master	Requires 8-bit data bus width.
SD1 (2.0)	0101	MIO[51:38]	Yes	Yes	Yes	Master	SD 2.0.
eMMC (1.8V)	0110	MIO[22:13]	Yes	Yes	Yes	Master	eMMC version 4.5 at 1.8V.
USB0 (2.0)	0111	MIO[52:63]	Yes	Yes	Yes	Slave	USB 2.0 only.
PJTAG (MIO #0)	1000	MIO[29:26]	Yes	No	No	Slave	PJTAG connection 0 option.
PJTAG (MIO #1)	1001	MIO[15:12]	Yes	No	No	Slave	PJTAG connection 1 option.
SD1 LS (3.0)	1110	MIO[51:39]	Yes	Yes	Yes	Master	SD 3.0 with a required SD 3.0 compliant voltage level shifter.

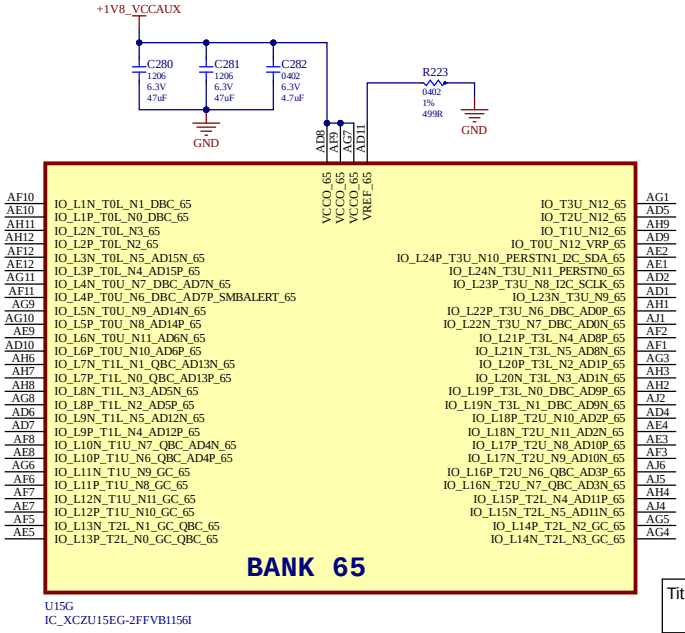
Title	5.2 BANK_502_503		
Approved By		Reviewed By	Drawn By
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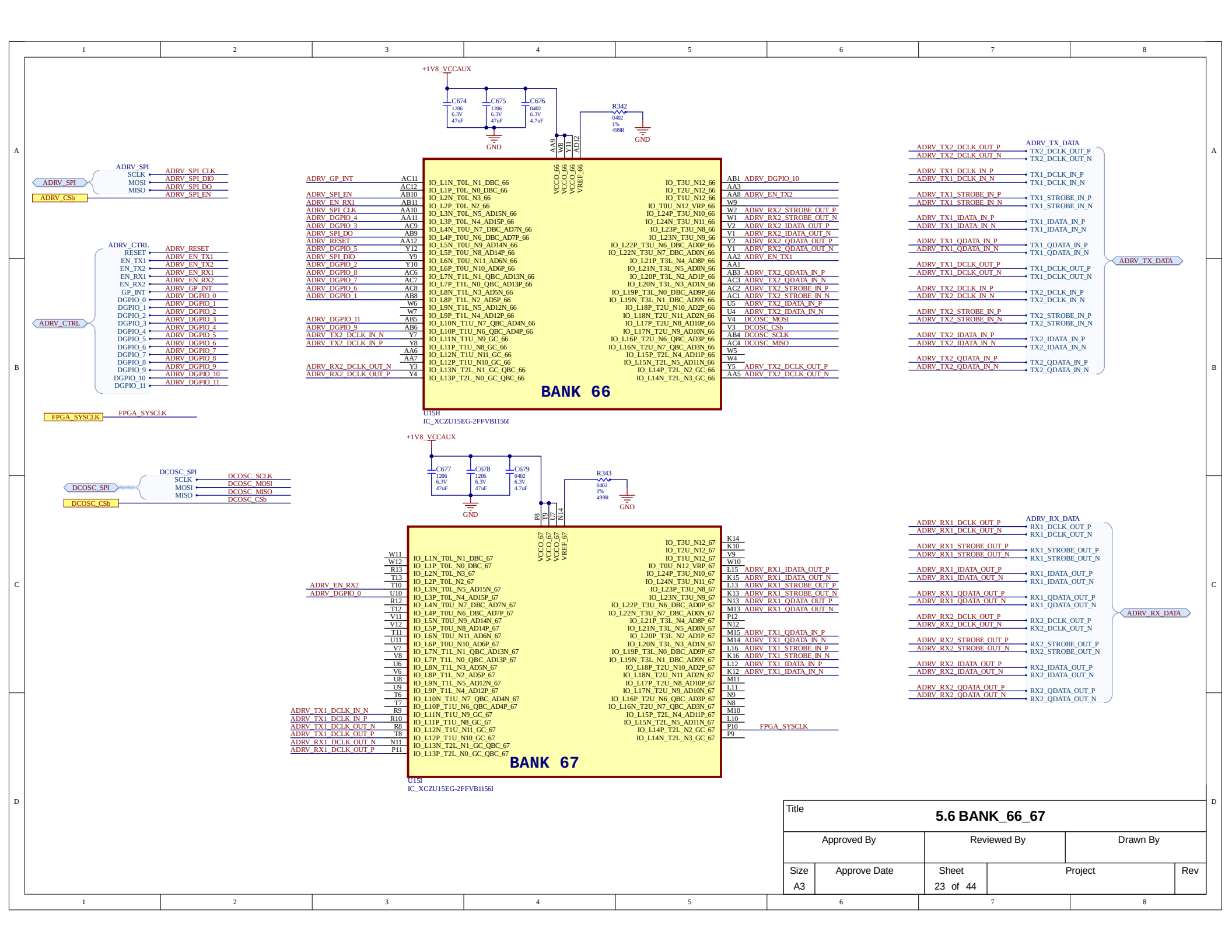
Title	5.3 BANK_504		
Approved By		Reviewed By	Drawn By
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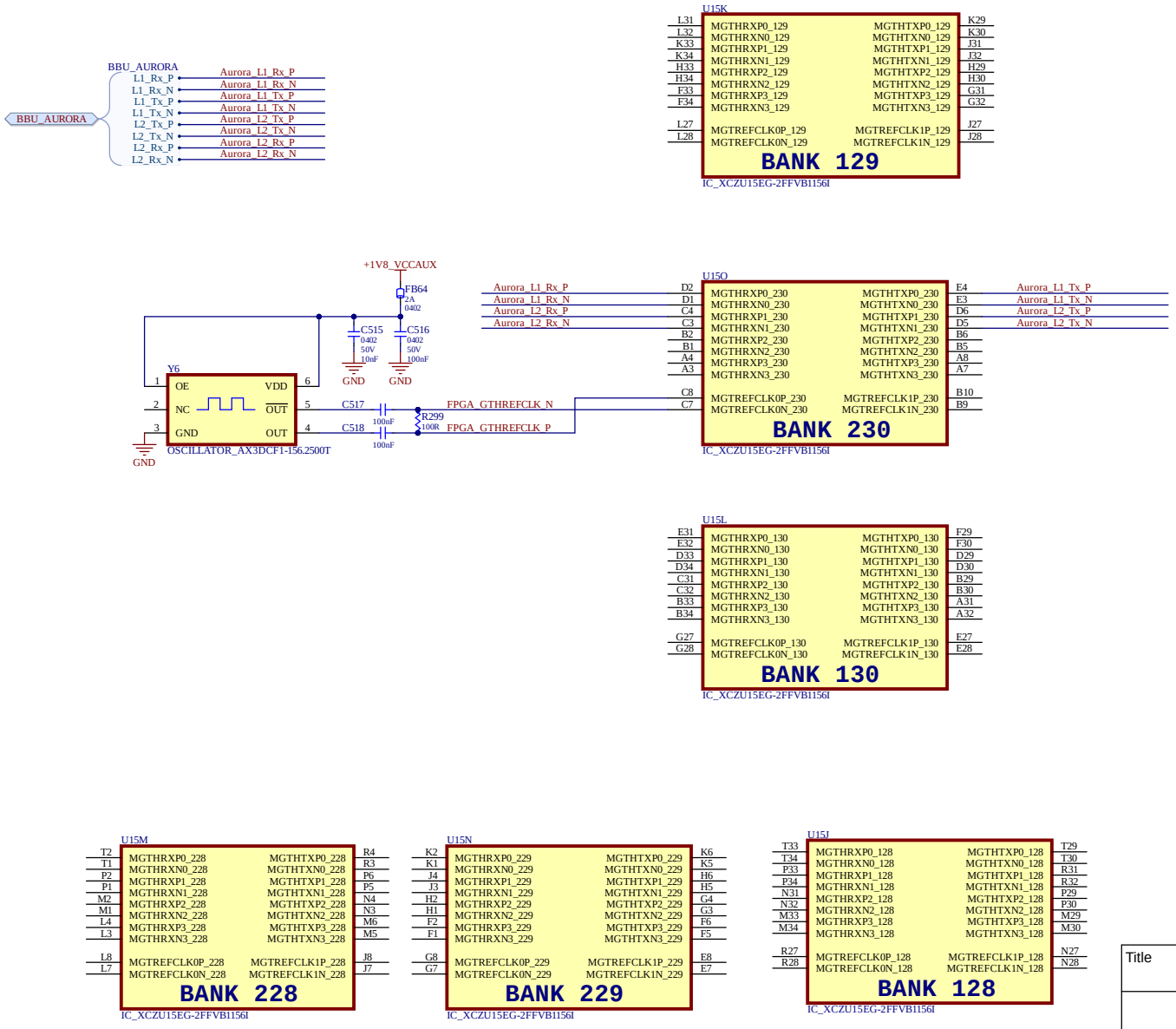

+3V3_VCCO
C267 0402 6.3V 10uF
C268 1206 6.3V 47uF
GND
AF14
AJ13
LB_TRX_CTRL AP14
LBF_CTRL3 AN14
HB_TRX_CTRL AN13
LBF_CTRL5 AM14
LBF_CTRL2 AP12
RFSW_CTRL3 AN12
HB_LNA_BYP AM13
LBF_CTRL4 AL13
TRX_MOSI AK14
LB_LNA_BYP AK15
RFSW_CTRL2 AL12
LBF_CTRL1 AK13
IO_L1N_AD11N_44
IO_L1P_AD11P_44
IO_L2N_AD10N_44
IO_L2P_AD10P_44
IO_L3N_AD9N_44
IO_L3P_AD9P_44
IO_L4N_AD8N_44
IO_L4P_AD8P_44
IO_L5N_HDGC_AD7N_44
IO_L5P_HDGC_AD7P_44
IO_L6N_HDGC_AD6N_44
IO_L6P_HDGC_AD6P_44
VCCO_44
VCCO_44
IO_L12P_AD0P_44
IO_L12N_AD0N_44
IO_L11P_AD1P_44
IO_L11N_AD1N_44
IO_L10P_AD2P_44
IO_L10N_AD2N_44
IO_L9P_AD3P_44
IO_L9N_AD3N_44
IO_L8P_HDGC_AD4P_44
IO_L8N_HDGC_AD4N_44
IO_L7P_HDGC_AD5P_44
IO_L7N_HDGC_AD5N_44
AE15 LED_ID1
AE14 LED_ID2
AF15 BBU_GPIO_2 (UART RX)
AG15 BBU_GPIO_3 (UART TX)
AG14
AG13 RFSW_CTRL1
AE13
AF13 TRX_MISO
AJ15 RFSW_CTRL4
AJ14 TRX_SCLK
AH14 BBU_GPIO_1 (10 MHz)
AH13 BBU_GPIO_0 (GPIO_0)
BANK 44
U15A
IC_XCZU15EG-2FFVB1156I
RFSW_CTRL
TFT_CTRL1_A_3V3 TFT_CTRL1_A
TFT_CTRL1_B_3V3 TFT_CTRL1_B
TFT_CTRL2_A_3V3 TFT_CTRL2_A
TFT_CTRL2_B_3V3 TFT_CTRL2_B
TFT_CTRL3_A_3V3 TFT_CTRL3_A
TFT_CTRL3_B_3V3 TFT_CTRL3_B
TFT_CTRL4_A_3V3 TFT_CTRL4_A
TFT_CTRL4_B_3V3 TFT_CTRL4_B
TFT_CTRL5_A_3V3 TFT_CTRL5_A
TFT_CTRL5_B_3V3 TFT_CTRL5_B
TFT_CTRL6_A_3V3 TFT_CTRL6_A
TFT_CTRL6_B_3V3 TFT_CTRL6_B
TFT_CTRL7_A_3V3 TFT_CTRL7_A
TFT_CTRL7_B_3V3 TFT_CTRL7_B
TFT_CTRL8_A_3V3 TFT_CTRL8_A
TFT_CTRL8_B_3V3 TFT_CTRL8_B
SW13_CTRL_A_3V3 SW13_CTRL_A
SW13_CTRL_B_3V3 SW13_CTRL_B
SW14_CTRL_A_3V3 SW14_CTRL_A
SW14_CTRL_B_3V3 SW14_CTRL_B
SW15_CTRL_A_3V3 SW15_CTRL_A
SW15_CTRL_B_3V3 SW15_CTRL_B
SW16_CTRL_A_3V3 SW16_CTRL_A
SW16_CTRL_B_3V3 SW16_CTRL_B
RFSW_CTRL_3V3
GB_LB_BYP_CTRL GB_LB_BYP_CTRL
GB1_HB_BYP_CTRL_3V3 GB1_HB_BYP_CTRL_3V3
GB2_HB_BYP_CTRL_3V3 GB2_HB_BYP_CTRL_3V3
HB_TFT_RESET HB_TFT_RESET
LB_TFT_RESET LB_TFT_RESET
LB_TFT_CSb1 LB_TFT_CSb1
LB_TFT_CSb2 LB_TFT_CSb2
HB_TFT_CSb1 HB_TFT_CSb1
HB_TFT_CSb2 HB_TFT_CSb2
HB_TFT_CSb3 HB_TFT_CSb3
HB_TFT_CSb4 HB_TFT_CSb4
RX_TEST_EN RX_TEST_EN
RX_GB_CTRL RX_GB_CTRL
TX_GB_CTRL1 TX_GB_CTRL1
TX_GB_CTRL2 TX_GB_CTRL2
TX_GB_CTRL3 TX_GB_CTRL3
LB_TFT_SPI
TFT_SPI
TFT_SCLK LB_TFT_SCLK
TFT_SDO LB_TFT_SDO
TFT_SDI LB_TFT_SDI
TFT_SFL LB_TFT_SFL
HB_TFT_SPI
TFT_SPI
TFT_SCLK HB_TFT_SCLK
TFT_SDO HB_TFT_SDO
TFT_SDI HB_TFT_SDI
TFT_SFL HB_TFT_SFL
ORX_DSA_SPI
DSA_SPI
SI ORX_DSA_SI
CLK ORX_DSA_CLK
LE ORX_DSA_LE
TX_DSA_SPI
DSA_SPI
SI TX_DSA_SI
CLK TX_DSA_CLK
LE TX_DSA_LE
RX_DSA_SPI
DSA_SPI
SI RX_DSA_SI
CLK RX_DSA_CLK
LE RX_DSA_LE
BBU_GPIO
BBU_GPIO
GPIO_0 (GPIO_0) BBU_GPIO_0 (GPIO_0)
GPIO_1 (10 MHz) BBU_GPIO_1 (10 MHz)
GPIO_2 (UART RX) BBU_GPIO_2 (UART RX)
GPIO_3 (UART TX) BBU_GPIO_3 (UART TX)
RFE_CTRL
RFE_CTRL
RFSW_CTRL1 RFSW_CTRL1
RFSW_CTRL2 RFSW_CTRL2
RFSW_CTRL3 RFSW_CTRL3
RFSW_CTRL4 RFSW_CTRL4
LBF_CTRL1 LBF_CTRL1
LBF_CTRL2 LBF_CTRL2
LBF_CTRL3 LBF_CTRL3
LBF_CTRL4 LBF_CTRL4
LBF_CTRL5 LBF_CTRL5
HB_LNA_BYP HB_LNA_BYP
LB_LNA_BYP LB_LNA_BYP
HB_TRX_CTRL HB_TRX_CTRL
LB_TRX_CTRL LB_TRX_CTRL
TRX_SCLK TRX_SCLK
TRX_MOSI TRX_MOSI
TRX_MISO TRX_MISO
LED_ID1 LED_ID1
LED_ID2 LED_ID2
+3V3_VCCO
C269 0402 6.3V 10uF
C270 1206 6.3V 47uF
GND
G17
J18
TFT_CTRL1_B_3V3 H19
SW13_CTRL_A_3V3 H18
GB2_HB_BYP_CTRL_3V3 H17
SW13_CTRL_B_3V3 J17
TFT_CTRL3_B_3V3 K18
TFT_CTRL1_A_3V3 L18
TX_GB_CTRL3 K17
TFT_CTRL3_A_3V3 L17
HB_TFT_CSb4 G19
HB_TFT_CSb1 G18
RX_DSA_CLK F18
TX_GB_CTRL2 F17
IO_L1N_AD15N_48
IO_L1P_AD15P_48
IO_L2N_AD14N_48
IO_L2P_AD14P_48
IO_L3N_AD13N_48
IO_L3P_AD13P_48
IO_L4N_AD12N_48
IO_L4P_AD12P_48
IO_L5N_HDGC_48
IO_L5P_HDGC_48
IO_L6N_HDGC_48
IO_L6P_HDGC_48
VCCO_48
VCCO_48
IO_L12P_AD8P_48
IO_L12N_AD8N_48
IO_L11P_AD9P_48
IO_L11N_AD9N_48
IO_L10P_AD10P_48
IO_L10N_AD10N_48
IO_L9P_AD11P_48
IO_L9N_AD11N_48
IO_L8P_HDGC_48
IO_L8N_HDGC_48
IO_L7P_HDGC_48
IO_L7N_HDGC_48
A17 TX_DSA_CLK
A18 TFT_CTRL7_A_3V3
C18 TX_DSA_SI
C19
B18 TX_DSA_LE
B19 SW16_CTRL_B_3V3
D17 RX_DSA_SI
C17
E17 RX_DSA_LE
E18 RX_GB_CTRL
E19 TX_GB_CTRL1
D19 SW16_CTRL_A_3V3
BANK 48
U15C
IC_XCZU15EG-2FFVB1156I
+3V3_VCCO
C271 0402 6.3V 10uF
C272 1206 6.3V 47uF
GND
E21
H20
TFT_CTRL7_B_3V3 K20
GB_LB_BYP_CTRL L20
TFT_CTRL2_B_3V3 K19
TFT_CTRL2_A_3V3 L19
TFT_CTRL8_A_3V3 H21
SW15_CTRL_A_3V3 J21
SW15_CTRL_B_3V3 J20
TFT_CTRL8_B_3V3 J19
SW14_CTRL_A_3V3 F21
GB1_HB_BYP_CTRL_3V3 G21
SW14_CTRL_B_3V3 F20
HB_TFT_RESET G20
IO_L1N_AD11N_47
IO_L1P_AD11P_47
IO_L2N_AD10N_47
IO_L2P_AD10P_47
IO_L3N_AD9N_47
IO_L3P_AD9P_47
IO_L4N_AD8N_47
IO_L4P_AD8P_47
IO_L5N_HDGC_AD7N_47
IO_L5P_HDGC_AD7P_47
IO_L6N_HDGC_AD6N_47
IO_L6P_HDGC_AD6P_47
VCCO_47
VCCO_47
IO_L12P_AD0P_47
IO_L12N_AD0N_47
IO_L11P_AD1P_47
IO_L11N_AD1N_47
IO_L10P_AD2P_47
IO_L10N_AD2N_47
IO_L9P_AD3P_47
IO_L9N_AD3N_47
IO_L8P_HDGC_AD4P_47
IO_L8N_HDGC_AD4N_47
IO_L7P_HDGC_AD5P_47
IO_L7N_HDGC_AD5N_47
B20 TFT_CTRL6_A_3V3
A20 TFT_CTRL6_B_3V3
A21 TFT_CTRL5_A_3V3
A22 HB_TFT_SDO
C21 HB_TFT_SCLK
B21 TFT_CTRL5_B_3V3
D21 HB_TFT_SDI
C22 HB_TFT_CSb3
E20 HB_TFT_CSb2
D20 HB_TFT_SFL
E22 TFT_CTRL4_A_3V3
D22 TFT_CTRL4_B_3V3
BANK 47
U15B
IC_XCZU15EG-2FFVB1156I
+3V3_VCCO
C275 0402 6.3V 10uF
C276 1206 6.3V 47uF
GND
G12
J13
J10 ORX_DSA_SI
J11 LB_TFT_SDO
G10 ORX_DSA_LE
H10 G16 LB_TFT_SFL
E10 J14 LB_TFT_SCLK
F10 H14 RX_TEST_EN
D10 G15 ORX_DSA_CLK
D11 G14 LB_TFT_SDI
G11 H13
H11 G13 LB_TFT_CSb2
F11 J12 LB_TFT_CSb1
F12 H12 LB_TFT_RESET
IO_L1N_AD15N_50
IO_L1P_AD15P_50
IO_L2N_AD14N_50
IO_L2P_AD14P_50
IO_L3N_AD13N_50
IO_L3P_AD13P_50
IO_L4N_AD12N_50
IO_L4P_AD12P_50
IO_L5N_HDGC_50
IO_L5P_HDGC_50
IO_L6N_HDGC_50
IO_L6P_HDGC_50
VCCO_50
VCCO_50
IO_L12P_AD8P_50
IO_L12N_AD8N_50
IO_L11P_AD9P_50
IO_L11N_AD9N_50
IO_L10P_AD10P_50
IO_L10N_AD10N_50
IO_L9P_AD11P_50
IO_L9N_AD11N_50
IO_L8P_HDGC_50
IO_L8N_HDGC_50
IO_L7P_HDGC_50
IO_L7N_HDGC_50
BANK 50
U15E
IC_XCZU15EG-2FFVB1156I
+3V3_VCCO
C273 0402 6.3V 10uF
C274 1206 6.3V 47uF
GND
E16
F14
F15
F16
C16
D16
A16
B16
A15
B15
D15
E15
D14
E14
IO_L1N_AD15N_49
IO_L1P_AD15P_49
IO_L2N_AD14N_49
IO_L2P_AD14P_49
IO_L3N_AD13N_49
IO_L3P_AD13P_49
IO_L4N_AD12N_49
IO_L4P_AD12P_49
IO_L5N_HDGC_49
IO_L5P_HDGC_49
IO_L6N_HDGC_49
IO_L6P_HDGC_49
VCCO_49
VCCO_49
IO_L12P_AD8P_49
IO_L12N_AD8N_49
IO_L11P_AD9P_49
IO_L11N_AD9N_49
IO_L10P_AD10P_49
IO_L10N_AD10N_49
IO_L9P_AD11P_49
IO_L9N_AD11N_49
IO_L8P_HDGC_49
IO_L8N_HDGC_49
IO_L7P_HDGC_49
IO_L7N_HDGC_49
F13
E13
E12
D12
C12
B12
A13
A12
C13
B13
C14
B14
BANK 49
U15D
IC_XCZU15EG-2FFVB1156I
Title
5.4 BANK_44_47_48_49_50
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Reviewed By
Drawn By
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+1V8_VCCAUX
C674 1206 6.3V 47uF
C675 1206 6.3V 47uF
C676 0402 6.3V 4.7uF
GND
R342 0402 1% 499R
GND
AA9 W8 Y11 AD12
VCCO_66 VCCO_66 VCCO_66 VREF_66
ADRV_SPI
SCLK MOSI MISO
ADRV_SPI_CLK ADRV_SPI_DIO ADRV_SPI_DO ADRV_SPI_EN
ADRV_CSb
ADRV_CTRL
RESET EN_TX1 EN_TX2 EN_RX1 EN_RX2 GP_INT DGPIO_0 DGPIO_1 DGPIO_2 DGPIO_3 DGPIO_4 DGPIO_5 DGPIO_6 DGPIO_7 DGPIO_8 DGPIO_9 DGPIO_10 DGPIO_11
ADRV_RESET ADRV_EN_TX1 ADRV_EN_TX2 ADRV_EN_RX1 ADRV_EN_RX2 ADRV_GP_INT ADRV_DGPIO_0 ADRV_DGPIO_1 ADRV_DGPIO_2 ADRV_DGPIO_3 ADRV_DGPIO_4 ADRV_DGPIO_5 ADRV_DGPIO_6 ADRV_DGPIO_7 ADRV_DGPIO_8 ADRV_DGPIO_9 ADRV_DGPIO_10 ADRV_DGPIO_11
FPGA_SYSCLK
FPGA_SYSCLK
ADRV_GP_INT AC11
ADRV_SPI_EN AB10
ADRV_EN_RX1 AB11
ADRV_SPI_CLK AA10
ADRV_DGPIO_4 AA11
ADRV_DGPIO_3 AC9
ADRV_SPI_DO AB9
ADRV_RESET AA12
ADRV_DGPIO_5 Y12
ADRV_SPI_DIO Y9
ADRV_DGPIO_2 Y10
ADRV_DGPIO_8 AC6
ADRV_DGPIO_7 AC7
ADRV_DGPIO_6 AC8
ADRV_DGPIO_1 AB8
W6
W7
ADRV_DGPIO_11 AB5
ADRV_DGPIO_9 AB6
ADRV_TX2_DCLK_IN_N Y7
ADRV_TX2_DCLK_IN_P Y8
AA6
AA7
ADRV_RX2_DCLK_OUT_N Y3
ADRV_RX2_DCLK_OUT_P Y4
IO_L1N_T0L_N1_DBC_66
IO_L1P_T0L_N0_DBC_66
IO_L2N_T0L_N3_66
IO_L2P_T0L_N2_66
IO_L3N_T0L_N5_AD15N_66
IO_L3P_T0L_N4_AD15P_66
IO_L4N_T0U_N7_DBC_AD7N_66
IO_L4P_T0U_N6_DBC_AD7P_66
IO_L5N_T0U_N9_AD14N_66
IO_L5P_T0U_N8_AD14P_66
IO_L6N_T0U_N11_AD6N_66
IO_L6P_T0U_N10_AD6P_66
IO_L7N_T1L_N1_QBC_AD13N_66
IO_L7P_T1L_N0_QBC_AD13P_66
IO_L8N_T1L_N3_AD5N_66
IO_L8P_T1L_N2_AD5P_66
IO_L9N_T1L_N5_AD12N_66
IO_L9P_T1L_N4_AD12P_66
IO_L10N_T1U_N7_QBC_AD4N_66
IO_L10P_T1U_N6_QBC_AD4P_66
IO_L11N_T1U_N9_GC_66
IO_L11P_T1U_N8_GC_66
IO_L12N_T1U_N11_GC_66
IO_L12P_T1U_N10_GC_66
IO_L13N_T2L_N1_GC_QBC_66
IO_L13P_T2L_N0_GC_QBC_66
IO_T3U_N12_66
IO_T2U_N12_66
IO_T1U_N12_66
IO_T0U_N12_VRP_66
IO_L24P_T3U_N10_66
IO_L24N_T3U_N11_66
IO_L23P_T3U_N8_66
IO_L23N_T3U_N9_66
IO_L22P_T3U_N6_DBC_AD0P_66
IO_L22N_T3U_N7_DBC_AD0N_66
IO_L21P_T3L_N4_AD8P_66
IO_L21N_T3L_N5_AD8N_66
IO_L20P_T3L_N2_AD1P_66
IO_L20N_T3L_N3_AD1N_66
IO_L19P_T3L_N0_DBC_AD9P_66
IO_L19N_T3L_N1_DBC_AD9N_66
IO_L18P_T2U_N10_AD2P_66
IO_L18N_T2U_N11_AD2N_66
IO_L17P_T2U_N8_AD10P_66
IO_L17N_T2U_N9_AD10N_66
IO_L16P_T2U_N6_QBC_AD3P_66
IO_L16N_T2U_N7_QBC_AD3N_66
IO_L15P_T2L_N4_AD11P_66
IO_L15N_T2L_N5_AD11N_66
IO_L14P_T2L_N2_GC_66
IO_L14N_T2L_N3_GC_66
AB1 ADRV_DGPIO_10
AA3
AA8 ADRV_EN_TX2
W9
W2 ADRV_RX2_STROBE_OUT_P
W1 ADRV_RX2_STROBE_OUT_N
V2 ADRV_RX2_IDATA_OUT_P
V1 ADRV_RX2_IDATA_OUT_N
Y2 ADRV_RX2_QDATA_OUT_P
Y1 ADRV_RX2_QDATA_OUT_N
AA2 ADRV_EN_TX1
AA1
AB3 ADRV_TX2_QDATA_IN_P
AC3 ADRV_TX2_QDATA_IN_N
AC2 ADRV_TX2_STROBE_IN_P
AC1 ADRV_TX2_STROBE_IN_N
U5 ADRV_TX2_IDATA_IN_P
U4 ADRV_TX2_IDATA_IN_N
V4 DCOSC_MOSI
V3 DCOSC_CSb
AB4 DCOSC_SCLK
AC4 DCOSC_MISO
W5
W4
Y5 ADRV_TX2_DCLK_OUT_P
AA5 ADRV_TX2_DCLK_OUT_N
BANK 66
U15H
IC_XCZU15EG-2FFVB1156I
ADRV_TX_DATA
ADRV_TX2_DCLK_OUT_P
ADRV_TX2_DCLK_OUT_N
ADRV_TX1_DCLK_IN_P
ADRV_TX1_DCLK_IN_N
ADRV_TX1_STROBE_IN_P
ADRV_TX1_STROBE_IN_N
ADRV_TX1_IDATA_IN_P
ADRV_TX1_IDATA_IN_N
ADRV_TX1_QDATA_IN_P
ADRV_TX1_QDATA_IN_N
ADRV_TX1_DCLK_OUT_P
ADRV_TX1_DCLK_OUT_N
ADRV_TX2_DCLK_IN_P
ADRV_TX2_DCLK_IN_N
ADRV_TX2_STROBE_IN_P
ADRV_TX2_STROBE_IN_N
ADRV_TX2_IDATA_IN_P
ADRV_TX2_IDATA_IN_N
ADRV_TX2_QDATA_IN_P
ADRV_TX2_QDATA_IN_N
TX2_DCLK_OUT_P
TX2_DCLK_OUT_N
TX1_DCLK_IN_P
TX1_DCLK_IN_N
TX1_STROBE_IN_P
TX1_STROBE_IN_N
TX1_IDATA_IN_P
TX1_IDATA_IN_N
TX1_QDATA_IN_P
TX1_QDATA_IN_N
TX1_DCLK_OUT_P
TX1_DCLK_OUT_N
TX2_DCLK_IN_P
TX2_DCLK_IN_N
TX2_STROBE_IN_P
TX2_STROBE_IN_N
TX2_IDATA_IN_P
TX2_IDATA_IN_N
TX2_QDATA_IN_P
TX2_QDATA_IN_N
+1V8_VCCAUX
C677 1206 6.3V 47uF
C678 1206 6.3V 47uF
C679 0402 6.3V 4.7uF
GND
R343 0402 1% 499R
GND
P8 T9 U7 N14
VCCO_67 VCCO_67 VCCO_67 VREF_67
DCOSC_SPI
SCLK MOSI MISO
DCOSC_SCLK DCOSC_MOSI DCOSC_MISO DCOSC_CSb
DCOSC_SPI
DCOSC_CSb
W11 IO_L1N_T0L_N1_DBC_67
W12 IO_L1P_T0L_N0_DBC_67
R13 IO_L2N_T0L_N3_67
T13 IO_L2P_T0L_N2_67
ADRV_EN_RX2 T10 IO_L3N_T0L_N5_AD15N_67
ADRV_DGPIO_0 U10 IO_L3P_T0L_N4_AD15P_67
R12 IO_L4N_T0U_N7_DBC_AD7N_67
T12 IO_L4P_T0U_N6_DBC_AD7P_67
V11 IO_L5N_T0U_N9_AD14N_67
V12 IO_L5P_T0U_N8_AD14P_67
T11 IO_L6N_T0U_N11_AD6N_67
U11 IO_L6P_T0U_N10_AD6P_67
V7 IO_L7N_T1L_N1_QBC_AD13N_67
V8 IO_L7P_T1L_N0_QBC_AD13P_67
U6 IO_L8N_T1L_N3_AD5N_67
V6 IO_L8P_T1L_N2_AD5P_67
U8 IO_L9N_T1L_N5_AD12N_67
U9 IO_L9P_T1L_N4_AD12P_67
T6 IO_L10N_T1U_N7_QBC_AD4N_67
T7 IO_L10P_T1U_N6_QBC_AD4P_67
ADRV_TX1_DCLK_IN_N R9 IO_L11N_T1U_N9_GC_67
ADRV_TX1_DCLK_IN_P R10 IO_L11P_T1U_N8_GC_67
ADRV_TX1_DCLK_OUT_N R8 IO_L12N_T1U_N11_GC_67
ADRV_TX1_DCLK_OUT_P T8 IO_L12P_T1U_N10_GC_67
ADRV_RX1_DCLK_OUT_N N11 IO_L13N_T2L_N1_GC_QBC_67
ADRV_RX1_DCLK_OUT_P P11 IO_L13P_T2L_N0_GC_QBC_67
IO_T3U_N12_67
IO_T2U_N12_67
IO_T1U_N12_67
IO_T0U_N12_VRP_67
IO_L24P_T3U_N10_67
IO_L24N_T3U_N11_67
IO_L23P_T3U_N8_67
IO_L23N_T3U_N9_67
IO_L22P_T3U_N6_DBC_AD0P_67
IO_L22N_T3U_N7_DBC_AD0N_67
IO_L21P_T3L_N4_AD8P_67
IO_L21N_T3L_N5_AD8N_67
IO_L20P_T3L_N2_AD1P_67
IO_L20N_T3L_N3_AD1N_67
IO_L19P_T3L_N0_DBC_AD9P_67
IO_L19N_T3L_N1_DBC_AD9N_67
IO_L18P_T2U_N10_AD2P_67
IO_L18N_T2U_N11_AD2N_67
IO_L17P_T2U_N8_AD10P_67
IO_L17N_T2U_N9_AD10N_67
IO_L16P_T2U_N6_QBC_AD3P_67
IO_L16N_T2U_N7_QBC_AD3N_67
IO_L15P_T2L_N4_AD11P_67
IO_L15N_T2L_N5_AD11N_67
IO_L14P_T2L_N2_GC_67
IO_L14N_T2L_N3_GC_67
K14
K10
V9
W10
L15 ADRV_RX1_IDATA_OUT_P
K15 ADRV_RX1_IDATA_OUT_N
L13 ADRV_RX1_STROBE_OUT_P
K13 ADRV_RX1_STROBE_OUT_N
N13 ADRV_RX1_QDATA_OUT_P
M13 ADRV_RX1_QDATA_OUT_N
P12
N12
M15 ADRV_TX1_QDATA_IN_P
M14 ADRV_TX1_QDATA_IN_N
L16 ADRV_TX1_STROBE_IN_P
K16 ADRV_TX1_STROBE_IN_N
L12 ADRV_TX1_IDATA_IN_P
K12 ADRV_TX1_IDATA_IN_N
M11
L11
N9
N8
M10
L10
P10 FPGA_SYSCLK
P9
BANK 67
U15I
IC_XCZU15EG-2FFVB1156I
ADRV_RX_DATA
ADRV_RX1_DCLK_OUT_P
ADRV_RX1_DCLK_OUT_N
ADRV_RX1_STROBE_OUT_P
ADRV_RX1_STROBE_OUT_N
ADRV_RX1_IDATA_OUT_P
ADRV_RX1_IDATA_OUT_N
ADRV_RX1_QDATA_OUT_P
ADRV_RX1_QDATA_OUT_N
ADRV_RX2_DCLK_OUT_P
ADRV_RX2_DCLK_OUT_N
ADRV_RX2_STROBE_OUT_P
ADRV_RX2_STROBE_OUT_N
ADRV_RX2_IDATA_OUT_P
ADRV_RX2_IDATA_OUT_N
ADRV_RX2_QDATA_OUT_P
ADRV_RX2_QDATA_OUT_N
RX1_DCLK_OUT_P
RX1_DCLK_OUT_N
RX1_STROBE_OUT_P
RX1_STROBE_OUT_N
RX1_IDATA_OUT_P
RX1_IDATA_OUT_N
RX1_QDATA_OUT_P
RX1_QDATA_OUT_N
RX2_DCLK_OUT_P
RX2_DCLK_OUT_N
RX2_STROBE_OUT_P
RX2_STROBE_OUT_N
RX2_IDATA_OUT_P
RX2_IDATA_OUT_N
RX2_QDATA_OUT_P
RX2_QDATA_OUT_N
Title
5.6 BANK_66_67
Approved By
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Title	5.7 BANK_GTH		
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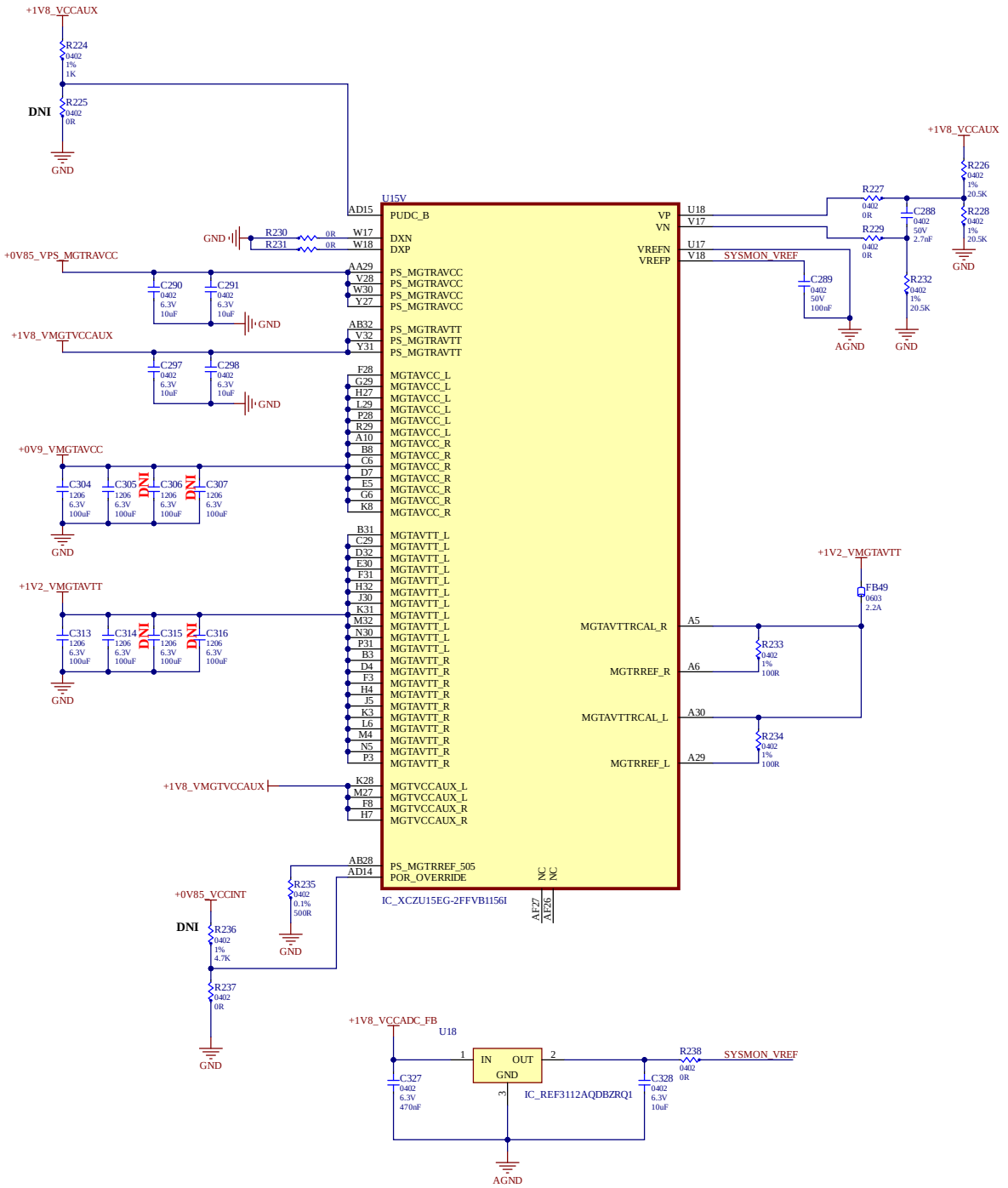


Table 5-8: **GTH Transceiver PCB Capacitor Recommendations**

Quantity Per Group			Capacitance (µF)	Tolerance	Type
MGTAVCC	MGTAVTT	MGTVCCAUX			
1	1	1	4.70	±10%	Ceramic

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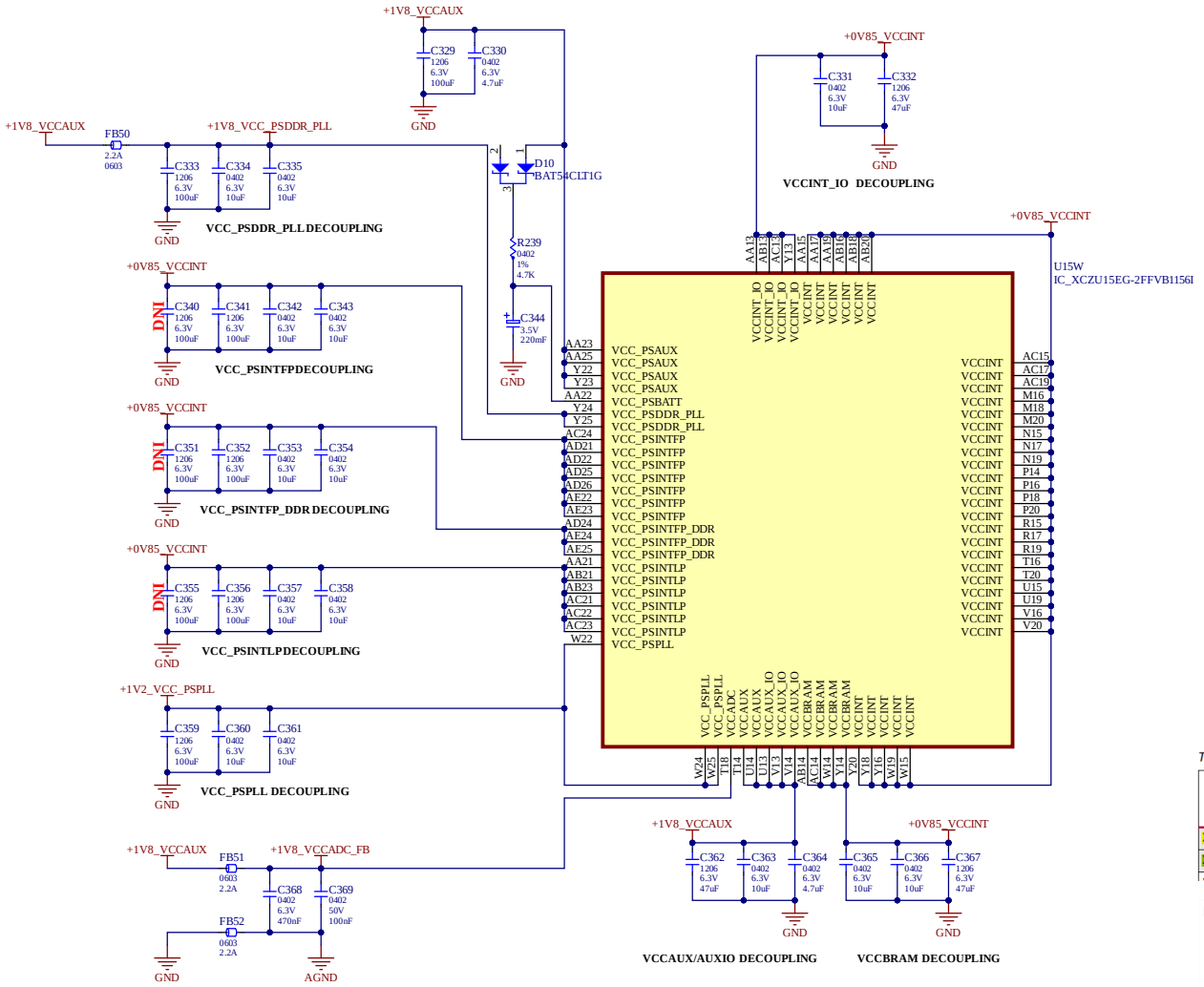


Table A-10: **Zynq UltraScale+ MPSoC PS Decoupling Capacitor Recommendations**

$V_{CC_PSINTFP}$		$V_{CC_PSINTLP}$		V_{CC_PSAUX}		V_{CC_PSPLL}		$V_{CC_PSINTFP_DDR}$		V_{CCO_PSIOx} [1] (Each)		V_{CC_PSBATT}	V_{CCO_PSDDR}	
100 µF	10 µF	100 µF	10 µF	100 µF	10 µF	100 µF	10 µF	100 µF	10 µF	100 µF	10 µF	10 µF	100 µF	10 µF
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Notes:

1. The 100 µF capacitor can be combined at one per every four shared V_{CCO_PSIO} banks.
2. For PS_MGTRAVCC and PS_MGTRAVTT use one 10 µF capacitor each. See Chapter 4, PCB Guidelines for the PS Interface in the Zynq UltraScale+ MPSoC.

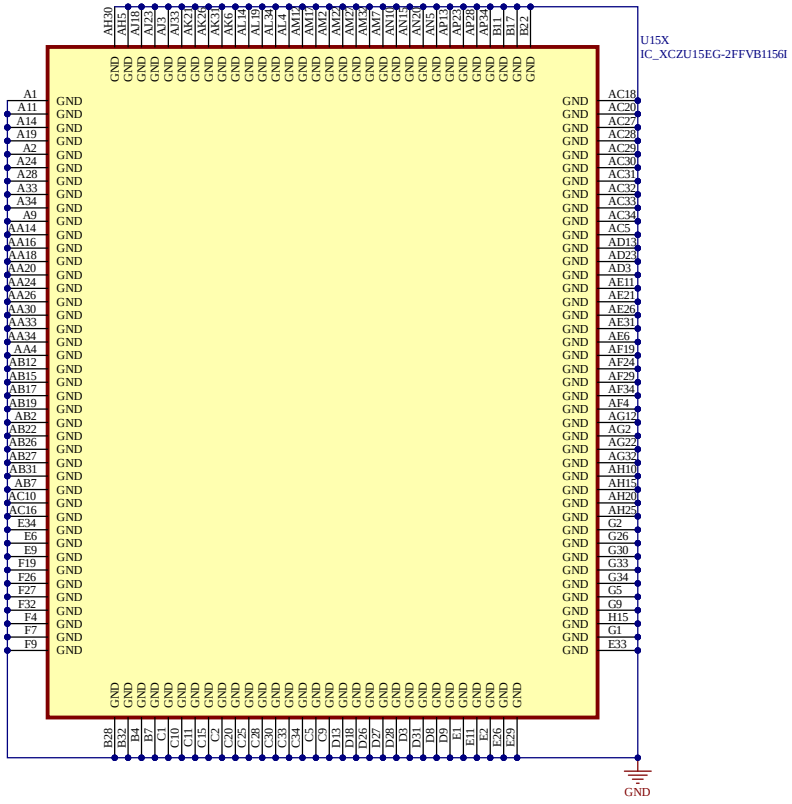
Table A-9: **Zynq UltraScale+ MPSoC Decoupling Capacitor Recommendations *(Cont'd)***

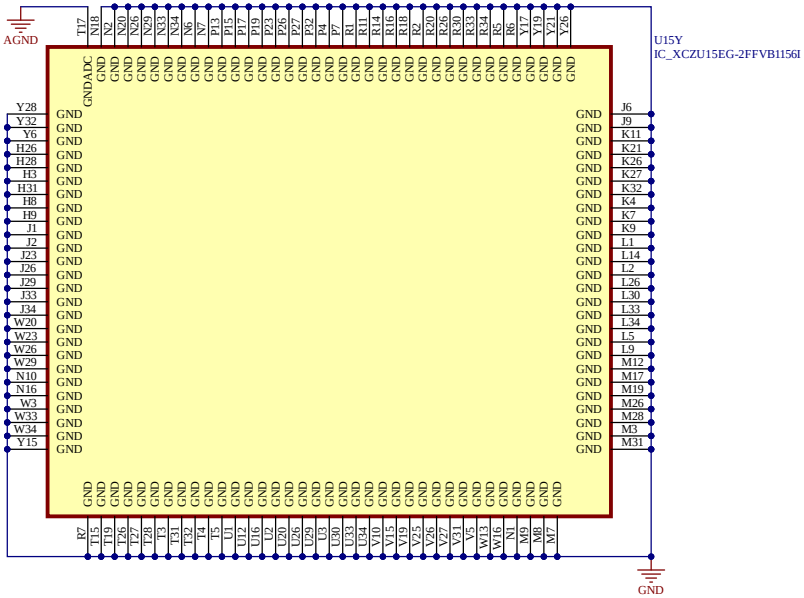
	V_{CCINT}/V_{CCINT_IO} [1]				V_{CCBRAM}/V_{CCINT_IO} [2]		V_{CCAUX}/V_{CCAUX_IO} [3]		HDIO/HPIO [4] (per bank)	
	330 µF	100 µF	47 µF	10 µF	47 µF	10 µF	47 µF	10 µF	47 µF	10 µF
XCZU15EG-FFVC900	1	5	1	1	1	1	1	1	1	1
XCZU15EG-FFVB1156	1	5	1	1	1	1	1	1	1	1

Notes:

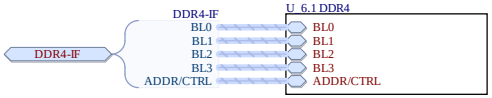
1. Connect V_{CCINT} and V_{CCINT_IO} together on the PCB for -3, -2, and -1 speed grades. The capacitors listed number of capacitors for the combined rail.
2. Connect V_{CCBRAM} and V_{CCINT_IO} together on the PCB for -2L and -1L speed grades. The capacitors listed number of capacitors for the combined rail.
3. V_{CCAUX} and V_{CCAUX_IO} must share the same plane on the PCB. The capacitors listed are the total number the combined rail.
4. The 47 µF capacitor can be combined at one per every four shared HDIO/HPIO banks.

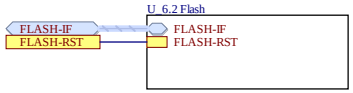
Title	**5.9 FPGA_PWR2**		
Approved By		Reviewed By	Drawn By
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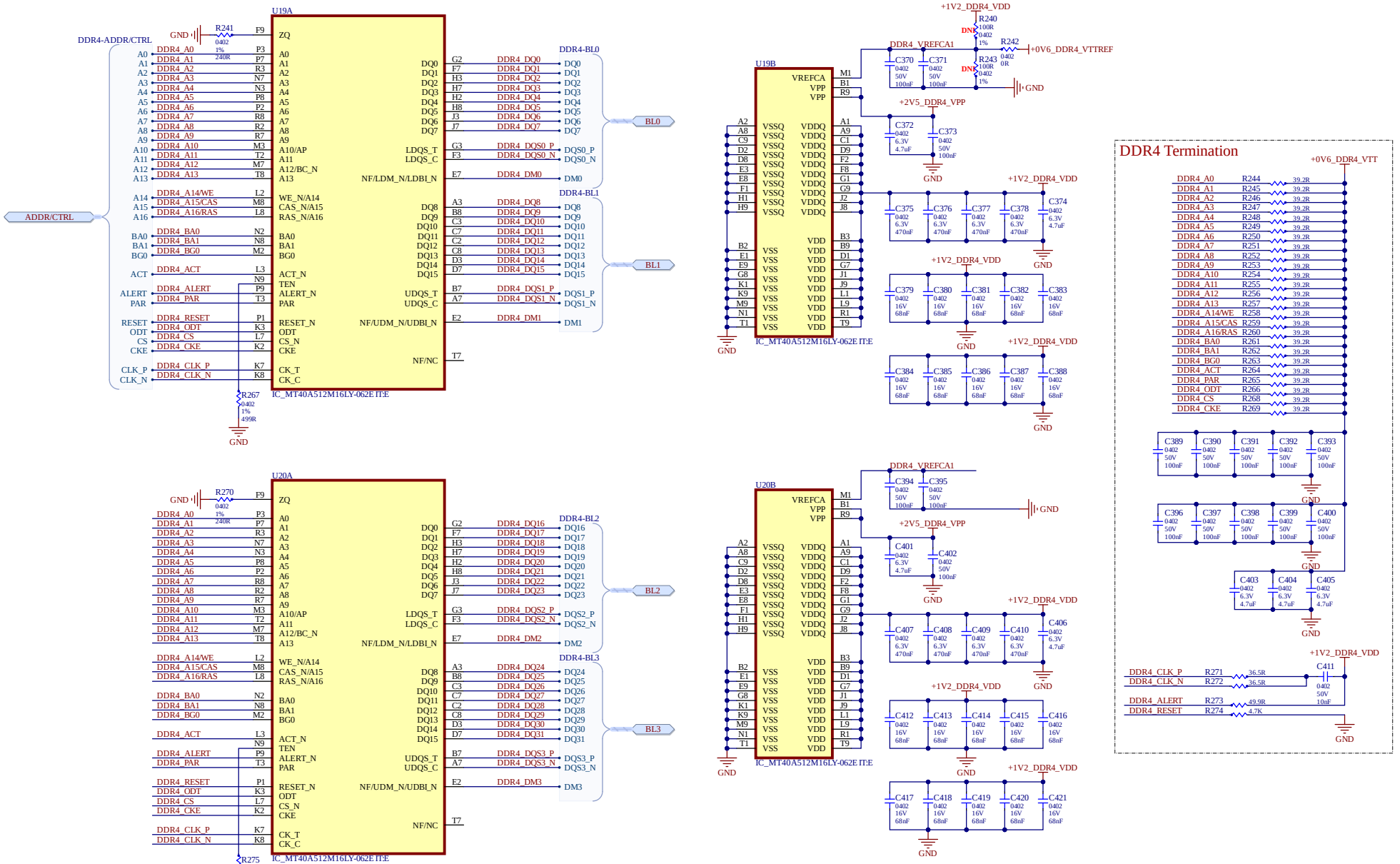


Title	**5.10 FPGA_GND**		
Approved By		Reviewed By	Drawn By
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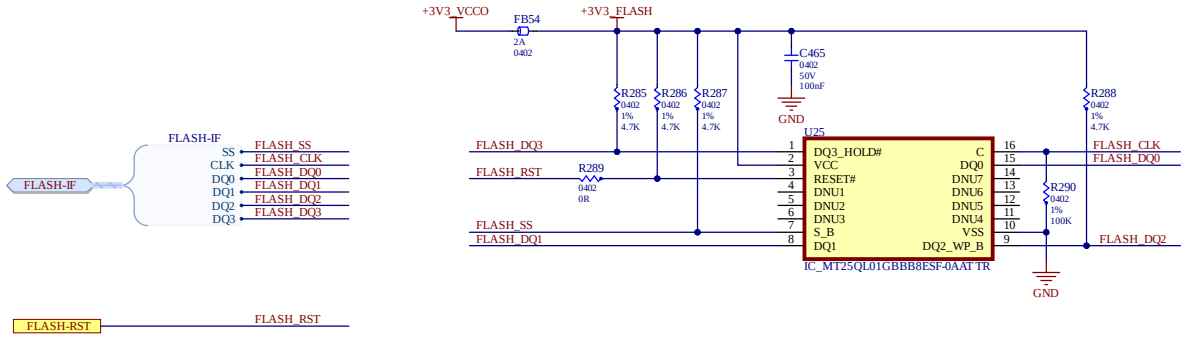




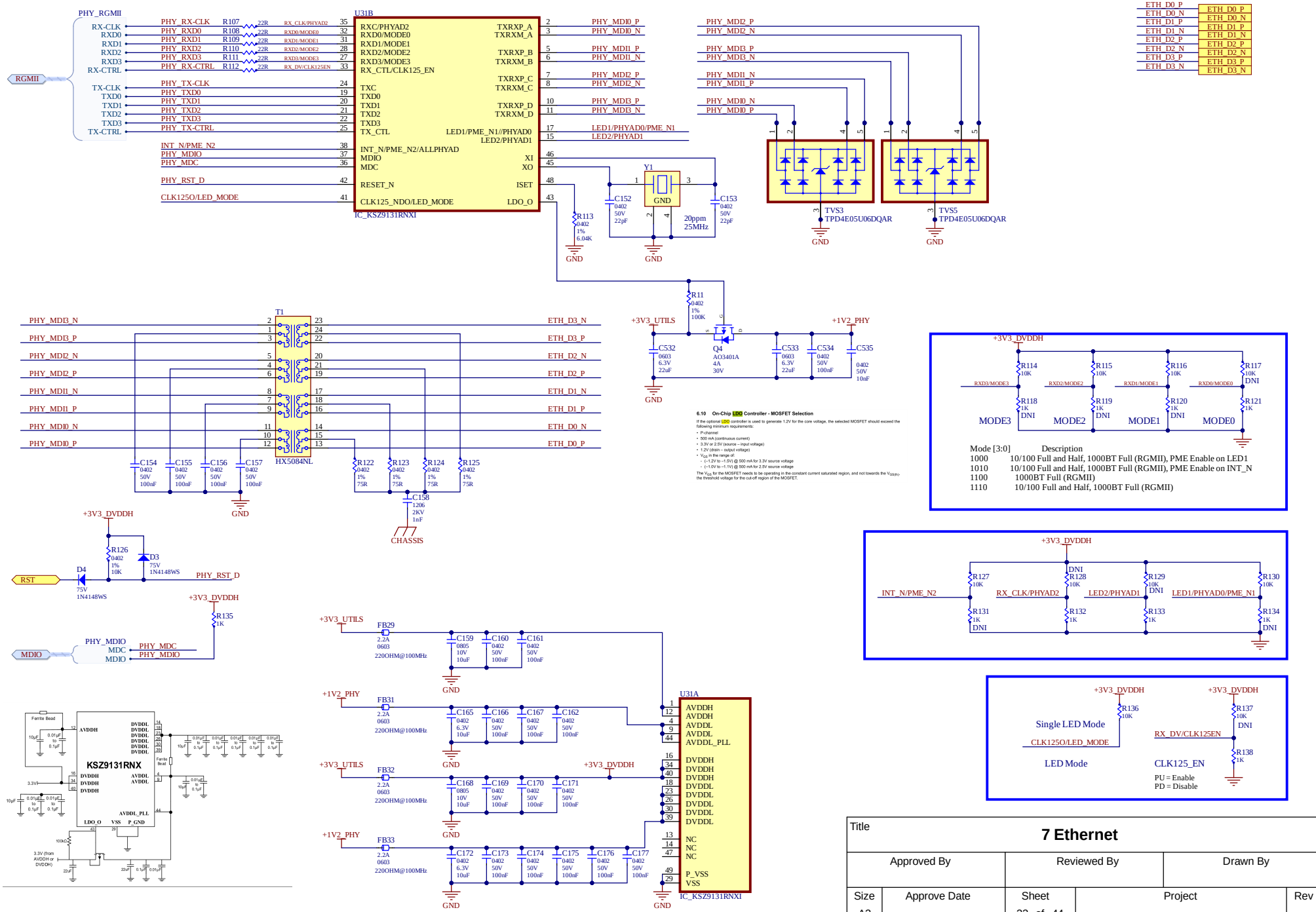
Title: **6 Memory**				
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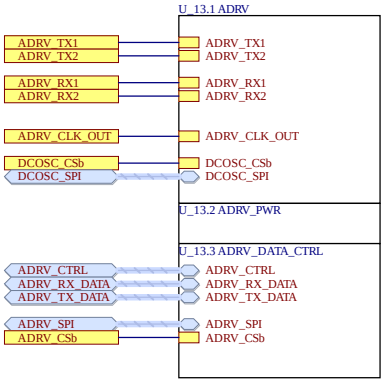


Title	6.1 DDR4_P01		
Approved By	Reviewed By	Drawn By	
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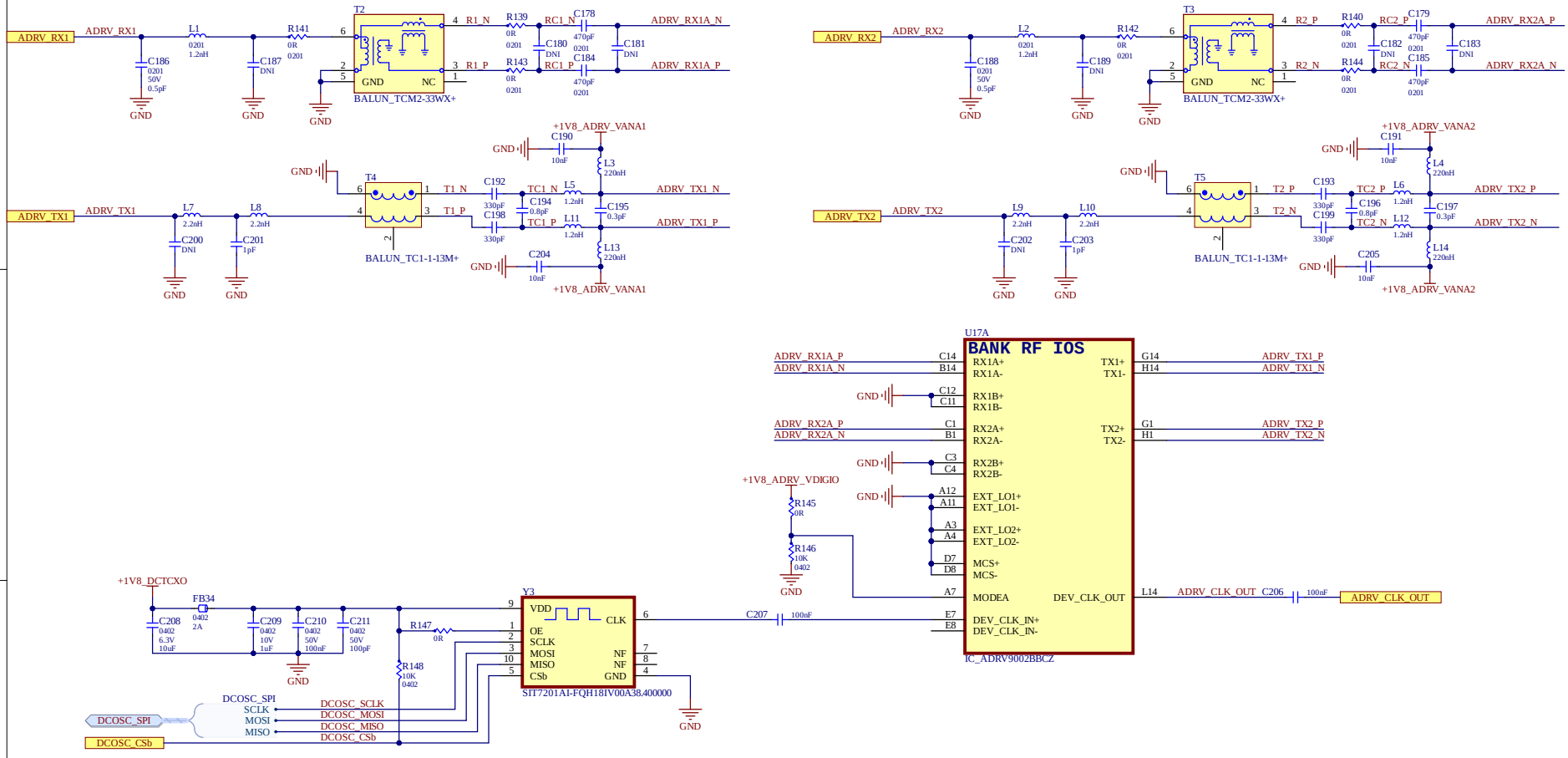


Title	**6.4 Flash_SDcard**		
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PHY_RGMII
RGMII
RX-CLK
RXD0
RXD1
RXD2
RXD3
RX-CTRL
TX-CLK
TXD0
TXD1
TXD2
TXD3
TX-CTRL
PHY_RX-CLK
PHY_RXD0
PHY_RXD1
PHY_RXD2
PHY_RXD3
PHY_RX-CTRL
PHY_TX-CLK
PHY_TXD0
PHY_TXD1
PHY_TXD2
PHY_TXD3
PHY_TX-CTRL
INT_N/PME_N2
PHY_MDIO
PHY_MDC
PHY_RST_D
CLK125O/LED_MODE
R107 22R RX_CLK/PHYAD2
R108 22R RXD0/MODE0
R109 22R RXD1/MODE1
R110 22R RXD2/MODE2
R111 22R RXD3/MODE3
R112 22R RX_DV/CLK125EN
U31B
RXC/PHYAD2
RXD0/MODE0
RXD1/MODE1
RXD2/MODE2
RXD3/MODE3
RX_CTL/CLK125_EN
TXC
TXD0
TXD1
TXD2
TXD3
TX_CTL
INT_N/PME_N2/ALLPHYAD
MDIO
MDC
RESET_N
CLK125_NDO/LED_MODE
TXRXP_A
TXRXM_A
TXRXP_B
TXRXM_B
TXRXP_C
TXRXM_C
TXRXP_D
TXRXM_D
LED1/PME_N1//PHYAD0
LED2/PHYAD1
XI
XO
ISET
LDO_O
IC_KSZ9131RNXI
PHY_MDI0_P
PHY_MDI0_N
PHY_MDI1_P
PHY_MDI1_N
PHY_MDI2_P
PHY_MDI2_N
PHY_MDI3_P
PHY_MDI3_N
LED1/PHYAD0/PME_N1
LED2/PHYAD1
Y1
GND
20ppm
25MHz
C152 0402 50V 22pF
C153 0402 50V 22pF
R113 0402 1% 6.04K
TVS3
TPD4E05U06DQAR
TVS5
TPD4E05U06DQAR
ETH_D0_P
ETH_D0_N
ETH_D1_P
ETH_D1_N
ETH_D2_P
ETH_D2_N
ETH_D3_P
ETH_D3_N
T1
HX5084NL
ETH_D3_N
ETH_D3_P
ETH_D2_N
ETH_D2_P
ETH_D1_N
ETH_D1_P
ETH_D0_N
ETH_D0_P
C154 0402 50V 100nF
C155 0402 50V 100nF
C156 0402 50V 100nF
C157 0402 50V 100nF
R122 0402 1% 75R
R123 0402 1% 75R
R124 0402 1% 75R
R125 0402 1% 75R
C158 1206 2KV 1nF
CHASSIS
+3V3_UTILS
+1V2_PHY
R11 0402 1% 100K
Q4 AO3401A 4A 30V
C532 0603 6.3V 22uF
C533 0603 6.3V 22uF
C534 0402 50V 100nF
C535 0402 50V 10nF
6.10 On-Chip LDO Controller - MOSFET Selection
+3V3_DVDDH
R114 10K
R115 10K
R116 10K
R117 10K DNI
R118 1K DNI
R119 1K DNI
R120 1K DNI
R121 1K
RXD3/MODE3
RXD2/MODE2
RXD1/MODE1
RXD0/MODE0
MODE3
MODE2
MODE1
MODE0
Mode [3:0] Description
1000 10/100 Full and Half, 1000BT Full (RGMII), PME Enable on LED1
1010 10/100 Full and Half, 1000BT Full (RGMII), PME Enable on INT_N
1100 1000BT Full (RGMII)
1110 10/100 Full and Half, 1000BT Full (RGMII)
R126 0402 1% 10K
D3 75V 1N4148WS
D4 75V 1N4148WS
RST
PHY_RST_D
R135 1K
MDIO
PHY_MDIO
PHY_MDC
INT_N/PME_N2
RX_CLK/PHYAD2
LED2/PHYAD1
LED1/PHYAD0/PME_N1
R127 10K
R128 10K DNI
R129 10K DNI
R130 10K
R131 1K DNI
R132 1K
R133 1K
R134 1K DNI
Single LED Mode
CLK125O/LED_MODE
LED Mode
R136 10K
R137 10K DNI
R138 1K
RX_DV/CLK125EN
CLK125_EN
PU = Enable
PD = Disable
FB29 2.2A 0603 220OHM@100MHz
FB31 2.2A 0603 220OHM@100MHz
FB32 2.2A 0603 220OHM@100MHz
FB33 2.2A 0603 220OHM@100MHz
C159 0805 10V 10uF
C160 0402 50V 100nF
C161 0402 50V 100nF
C165 0402 6.3V 10uF
C166 0402 50V 100nF
C167 0402 50V 100nF
C162 0402 50V 100nF
C168 0805 10V 10uF
C169 0402 50V 100nF
C170 0402 50V 100nF
C171 0402 50V 100nF
C172 0402 6.3V 10uF
C173 0402 50V 100nF
C174 0402 50V 100nF
C175 0402 50V 100nF
C176 0402 50V 100nF
C177 0402 50V 100nF
U31A
AVDDH
AVDDL
AVDDL_PLL
DVDDH
DVDDL
NC
P_VSS
VSS
IC_KSZ9131RNXI
KSZ9131RNX
Ferrite Bead
3.3V (from AVDDH or DVDDH)
Title
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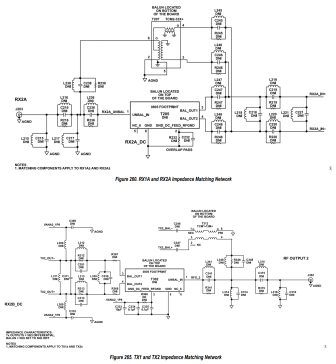


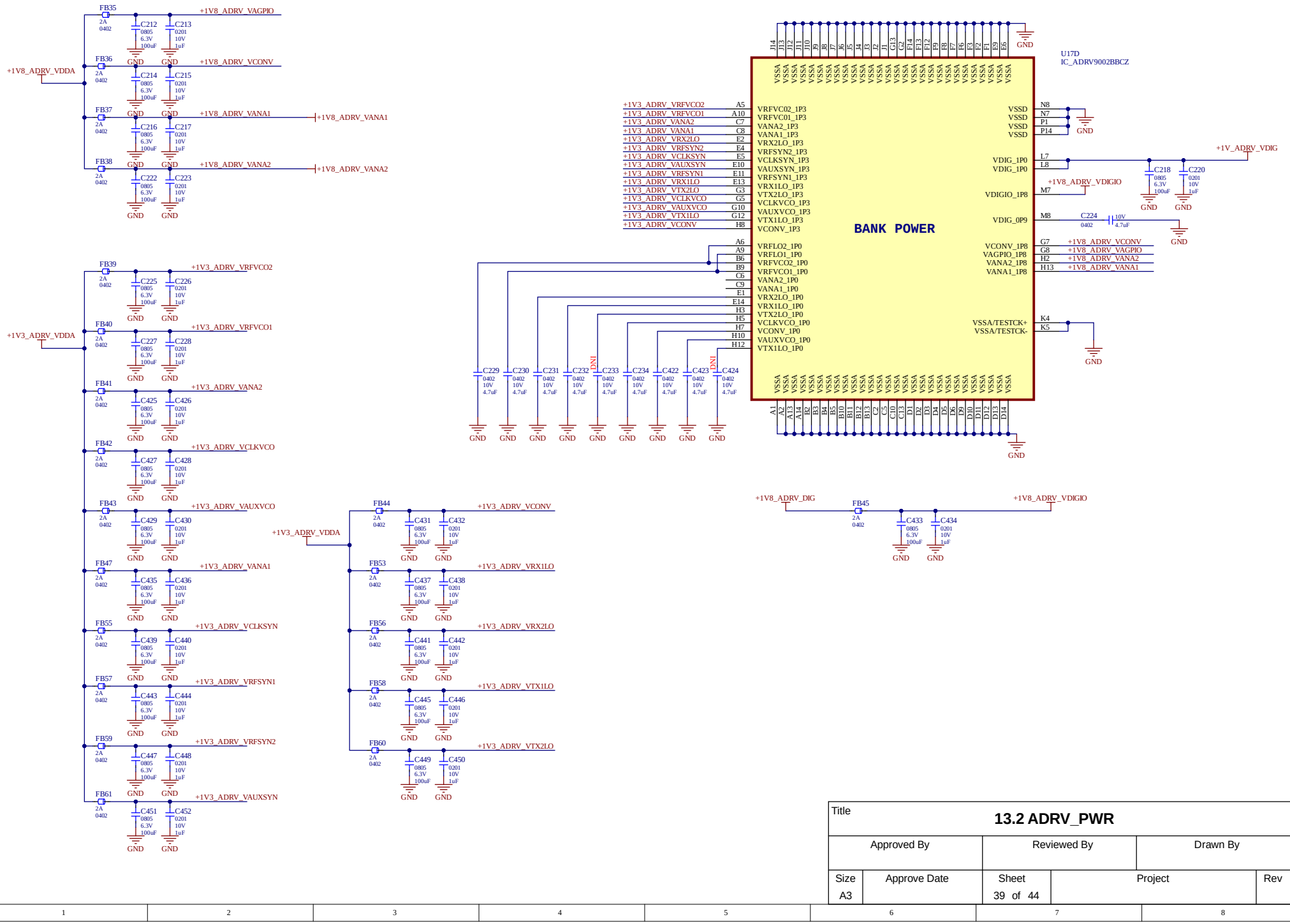
Table 112. RX1A± and RX2A± Impedance Matching Network

Frequency	Balun	L/C/R 215	L/C/R 216	L/C 217	L/C/R 238	L/C/R 236	L/C/R 245 L/C/R 246	L/C/R 247 L/C/R 248	L/C 218	L/C/R 219 L/C/R 220	L/C 221	C232 R232
30MHz to 3GHz	MiniCircuits TCM2-33WX+	L215: DNI C215: DNI	L216: 1.2nH AVX Ind (L0201 Series)	L217: DNI C217: DNI	L238: DNI C238: DNI R238: 0Ω	L236: DNI C236: DNI R236: DNI	L245/246: DNI C245/246: DNI R245/246: 0Ω	L247/248: DNI C247/248: DNI R247/248: DNI	L218: DNI C218: DNI	L219/220: DNI C219/220: 470 pF Murata	L221: DNI C221: DNI	C232:DNI R232:DNI

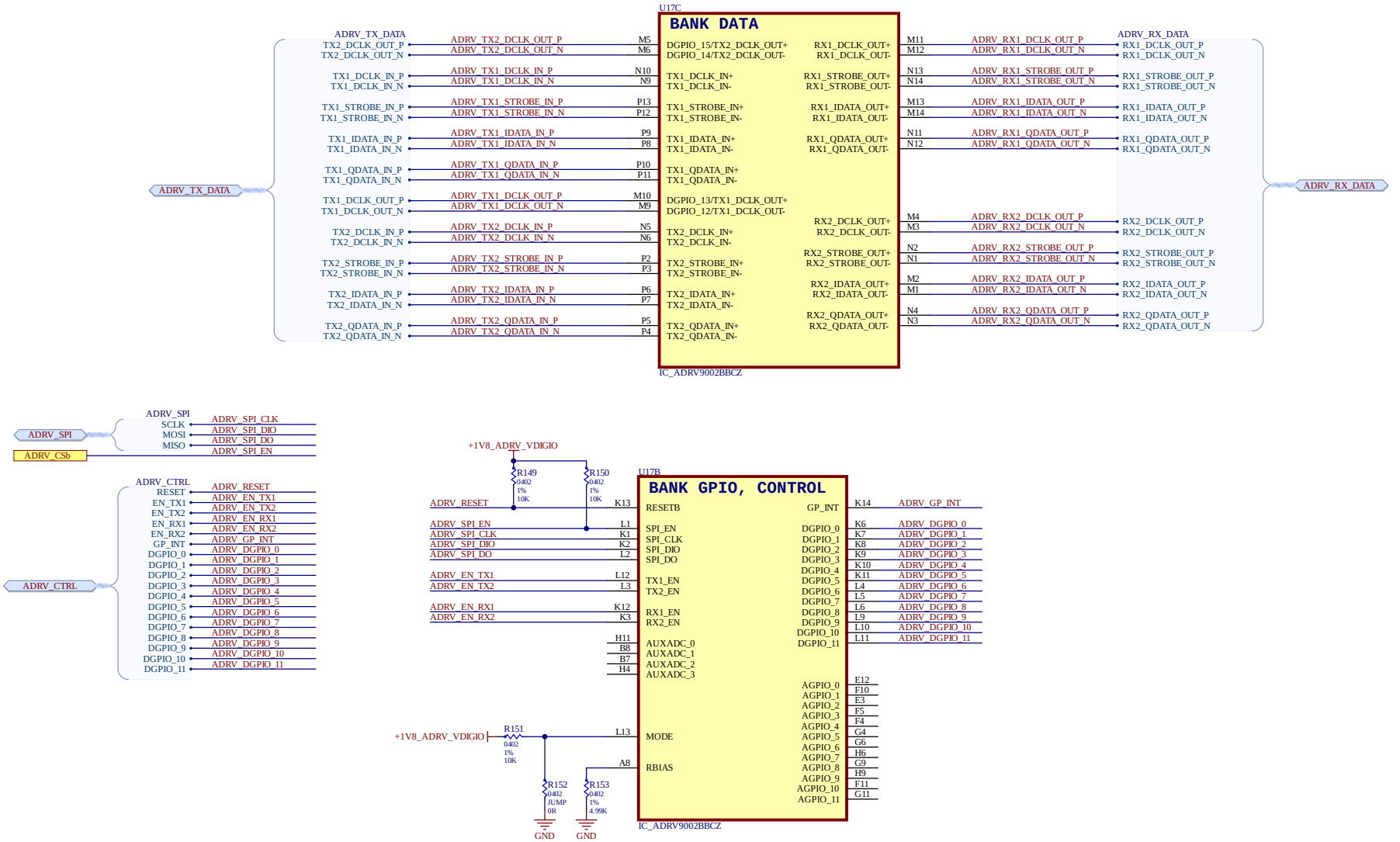
Frequency	Balun	L/C 311	L309 L310	C309 C310	L/C/R 312 L/C/R 313	L339 C339	C346/34 7 R307/36 8	R361	C333 C334 C335	L/C/R 341	L/C/R 348	L/C 314	L/C/R 315	L/C 316
30MHz – 3GHz	MiniCircuits TC-1-13M+	L311: DNI C311: 0.3pF (Murata GJM03)	L309/310: 220nH (CoilCraft LOW18AN)	C309 C310: 10nF (Murata GRM03)	L312/313: 1.2nH AVX Ind L0201 Series C312/313: DNI R312/313: DNI	L339: DNI C339: 0.8pF (Murata GJM03)	C346/34 7: 330pF (Murata GRM03) R307, R368: DNI	R361: DNI	C333: DNI C334: DNI C335: DNI	L341: DNI C341: DNI R341: DNI	L348: 2.2nH AVX Ind (L020 1 Series C348: DNI R348: DNI	L314: DNI C314: 1pF (Murata GJM03)	L315: 2.2nH AVX Ind (L0201 Series) C315: DNI R315: DNI	L316: DNI C316: DNI

If unused, connect MCS+ and MCS− to VSSA.
If unused, connect EXT_LO1+ and EXT_LO1− to VSSA.
If unused, connect EXT_LO2+ and EXT_LO2− to VSSA.
If unused, connect RX2B+ and RX2B− to VSSA.
If unused, connect RX1B− and RX1B+ to VSSA.
Connect MODEA to VSSA to enable the differential clock receiver at the DEV_CLK_IN± pins.

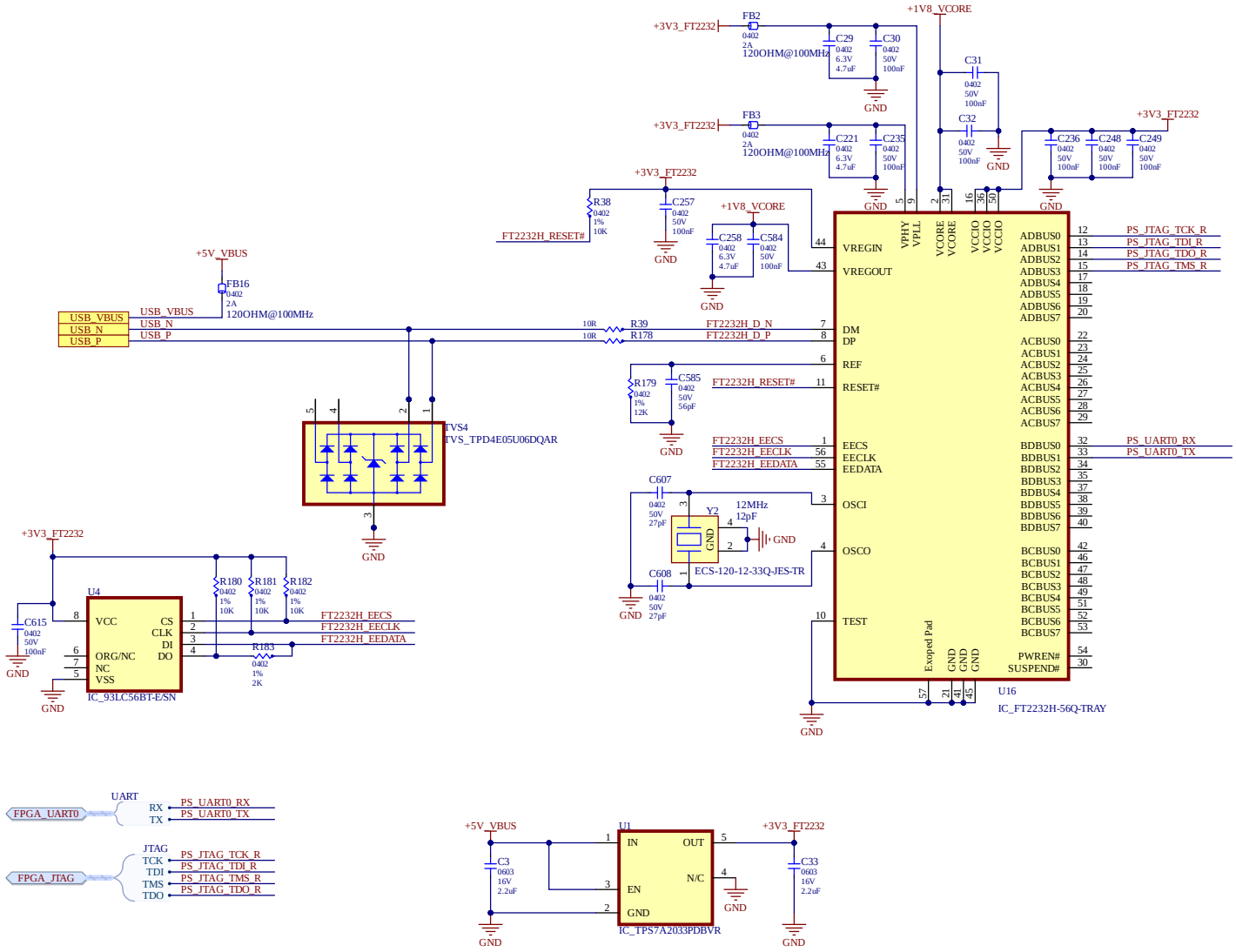
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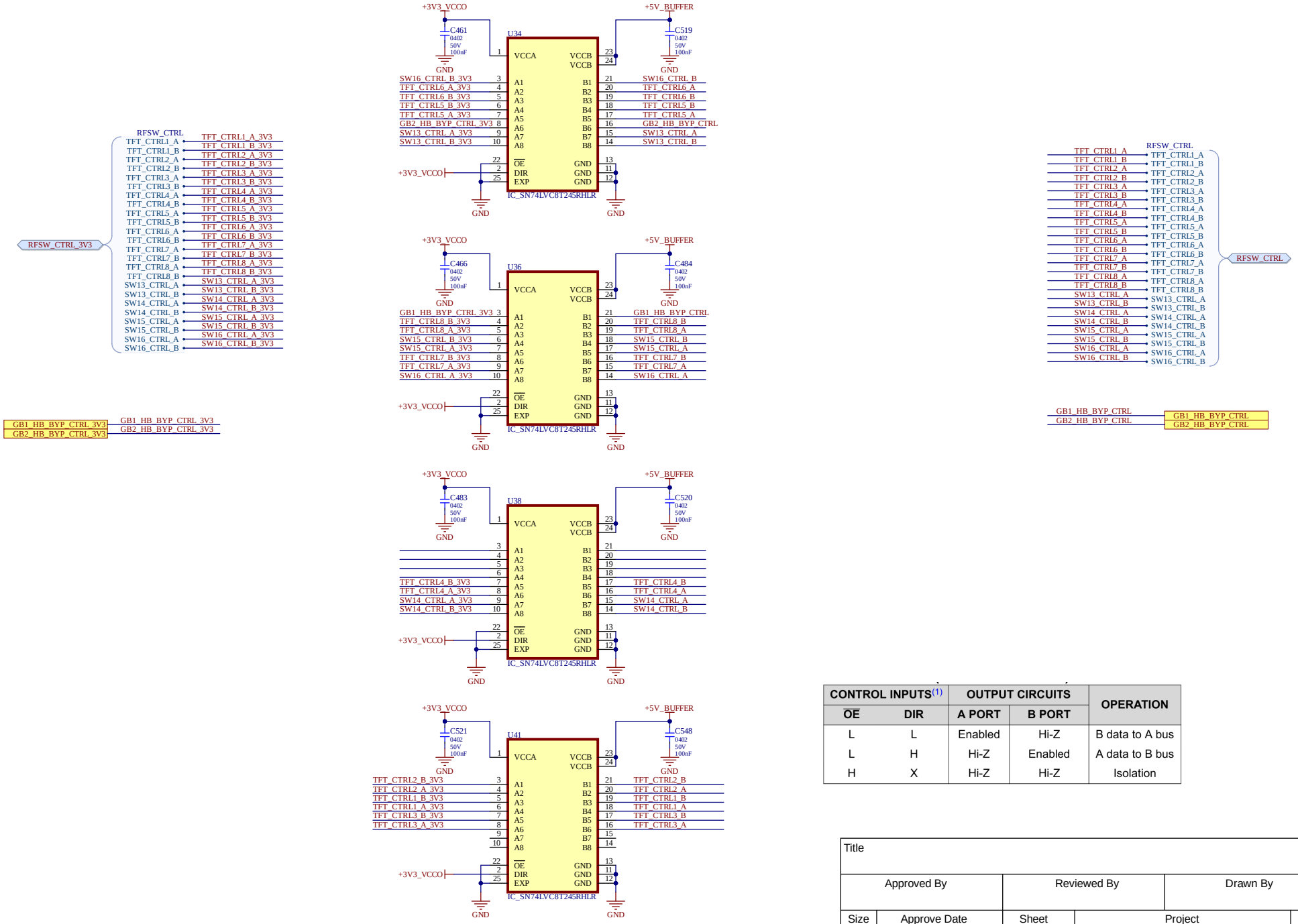
Title	13.2 ADRV_PWR		
Approved By		Reviewed By	Drawn By
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Title	13.3 RF_TX_Path		
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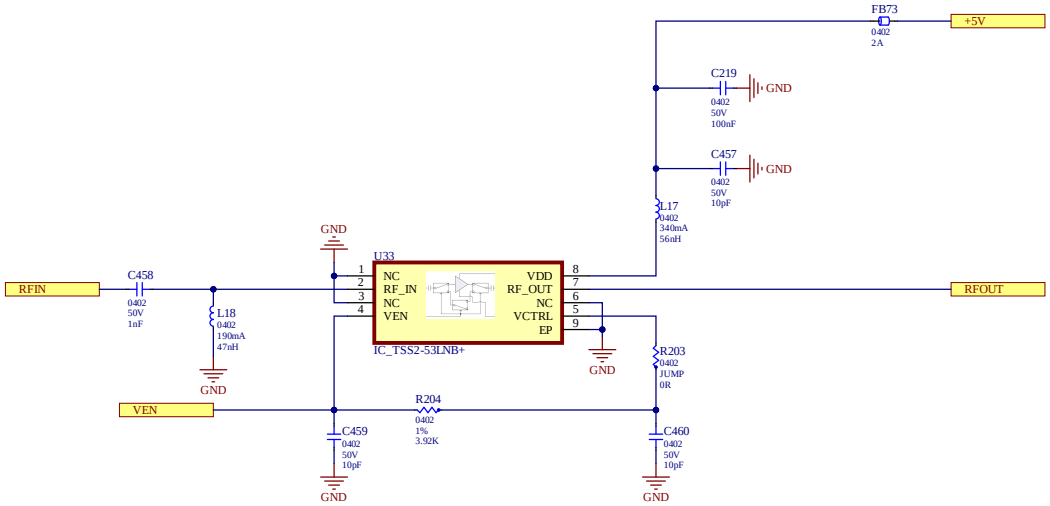


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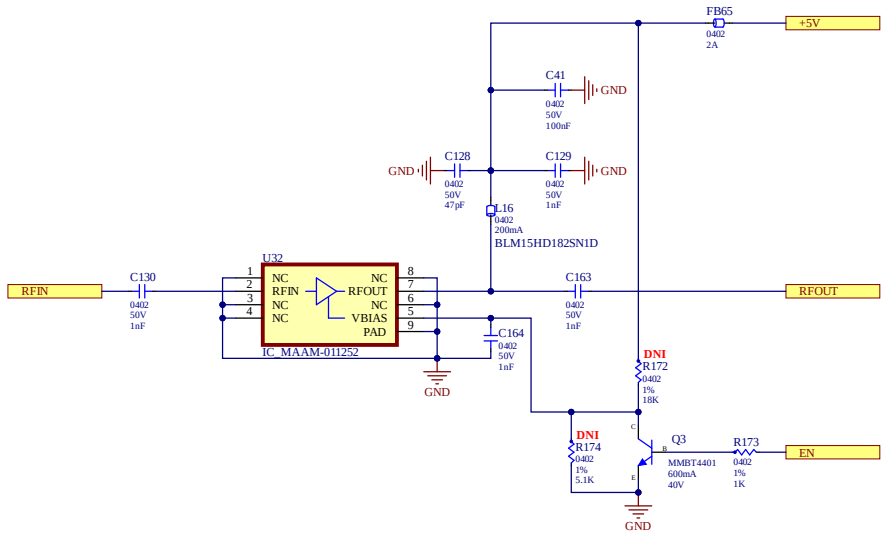


CONTROL INPUTS(1)		OUTPUT CIRCUITS		OPERATION
$\overline{OE}$	DIR	A PORT	B PORT	
L	L	Enabled	Hi-Z	B data to A bus
L	H	Hi-Z	Enabled	A data to B bus
H	X	Hi-Z	Hi-Z	Isolation

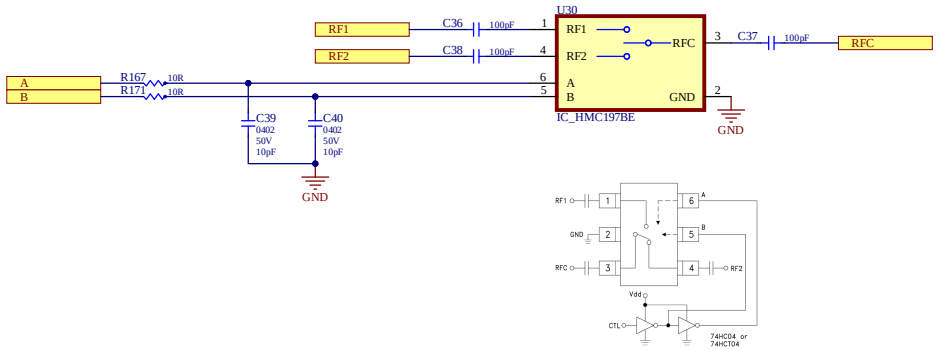
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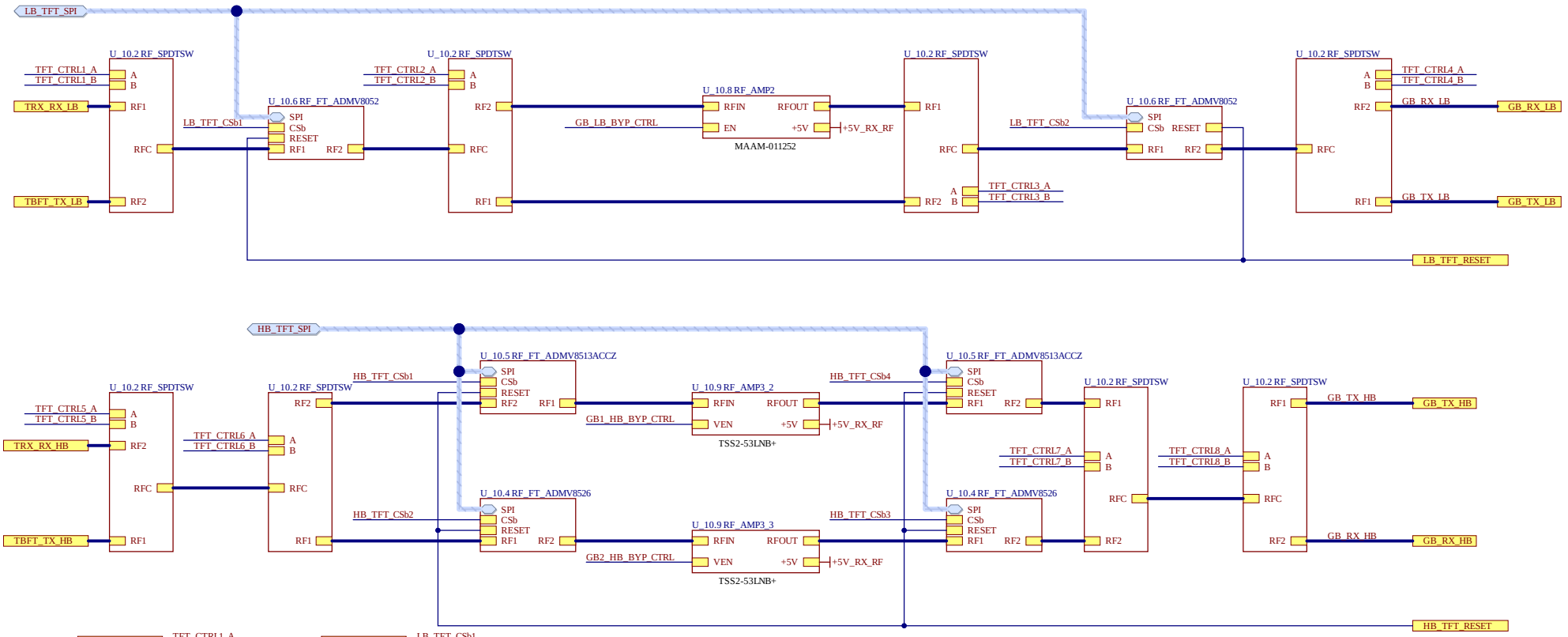
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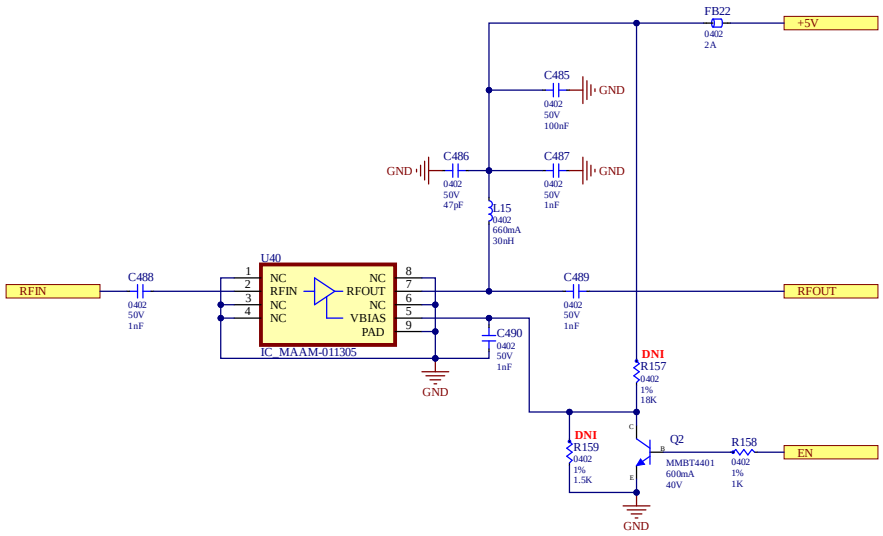


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TFT_CTRL6_B	TFT_CTRL6_B
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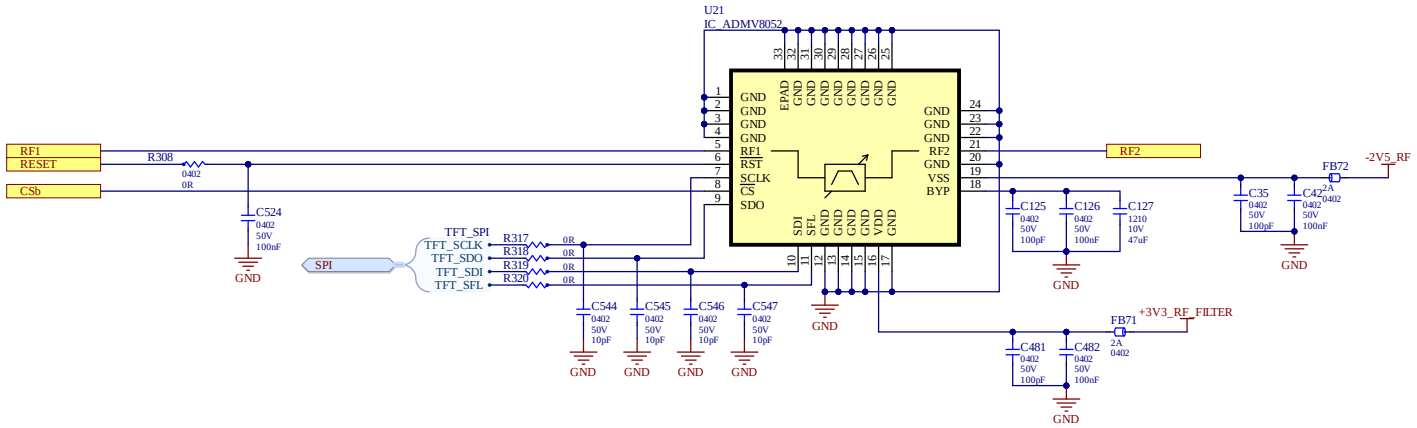
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HB_TFT_CSb3	HB_TFT_CSb3
HB_TFT_CSb4	HB_TFT_CSb4

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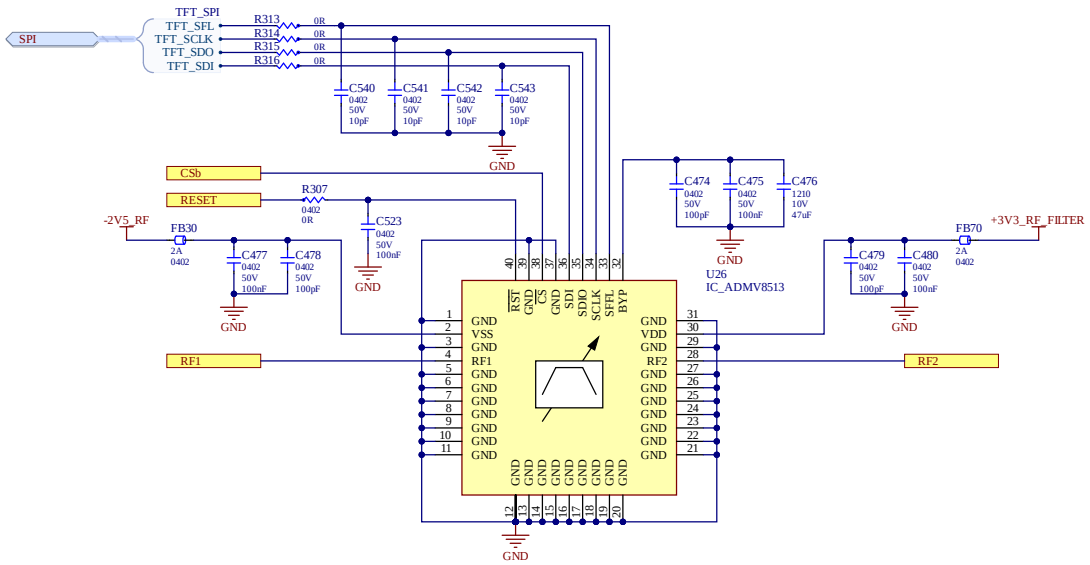
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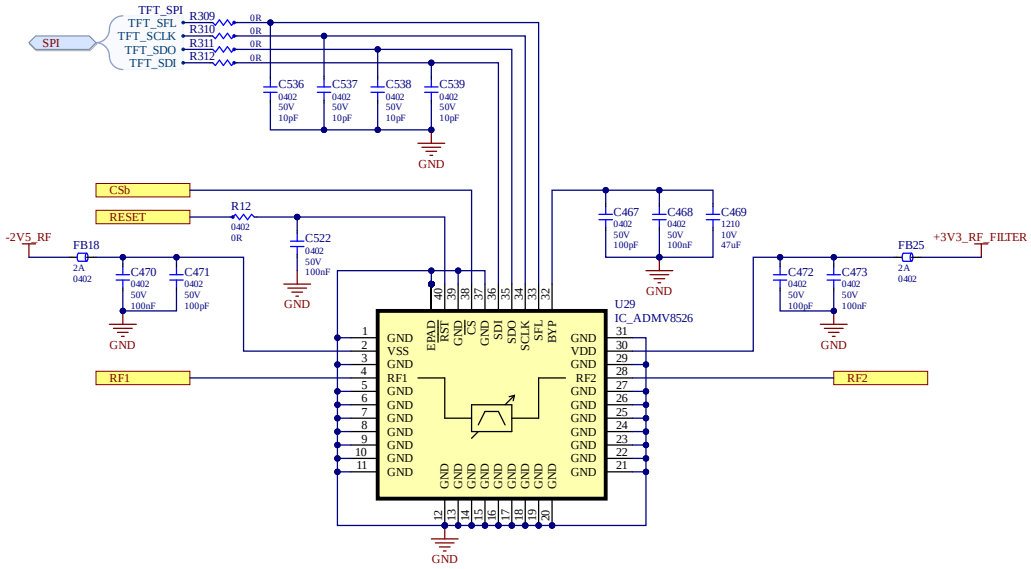
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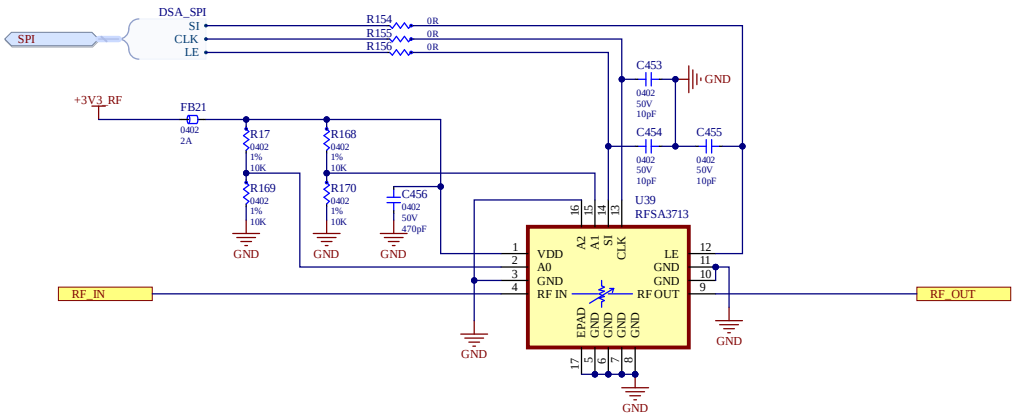
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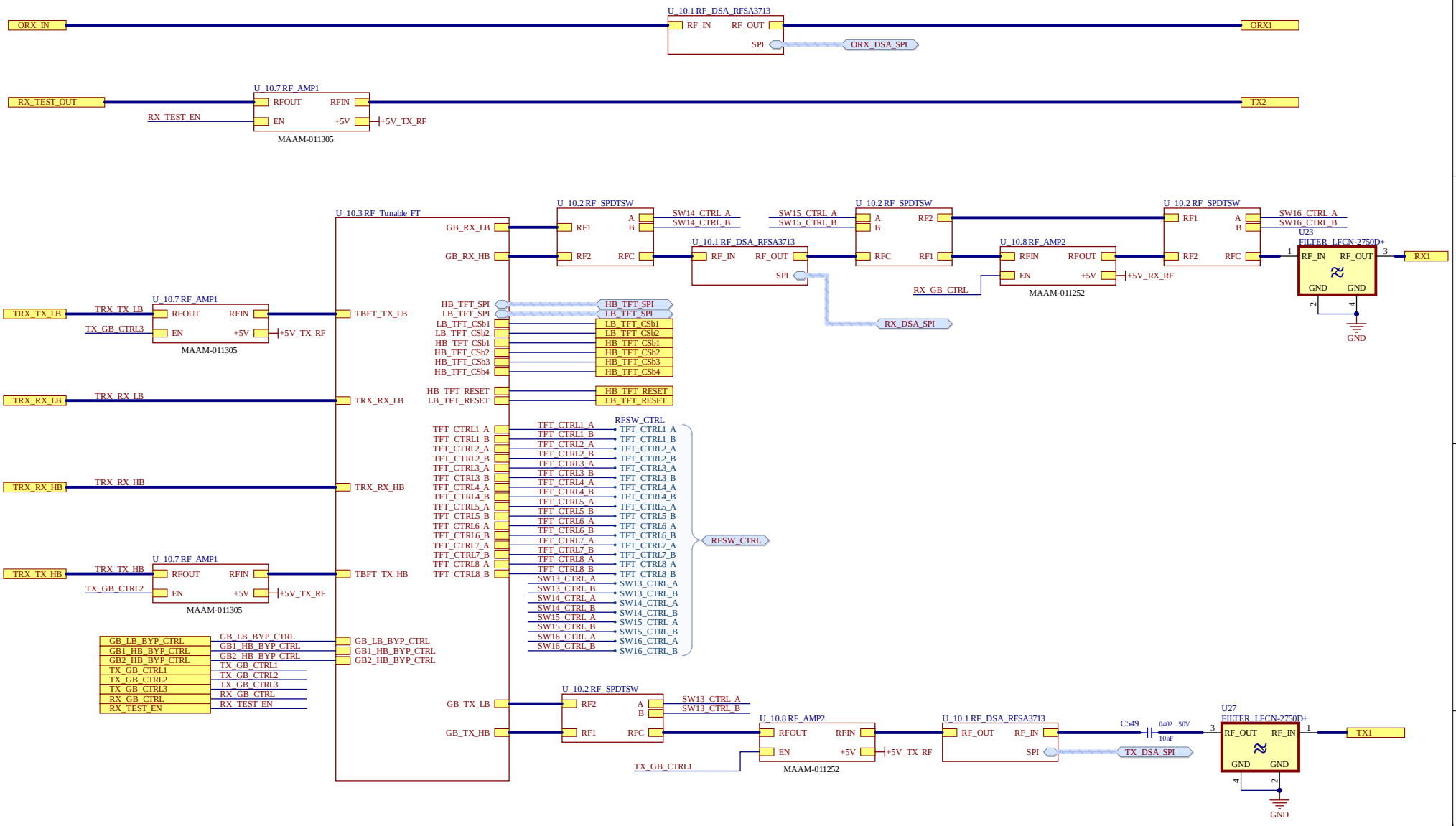
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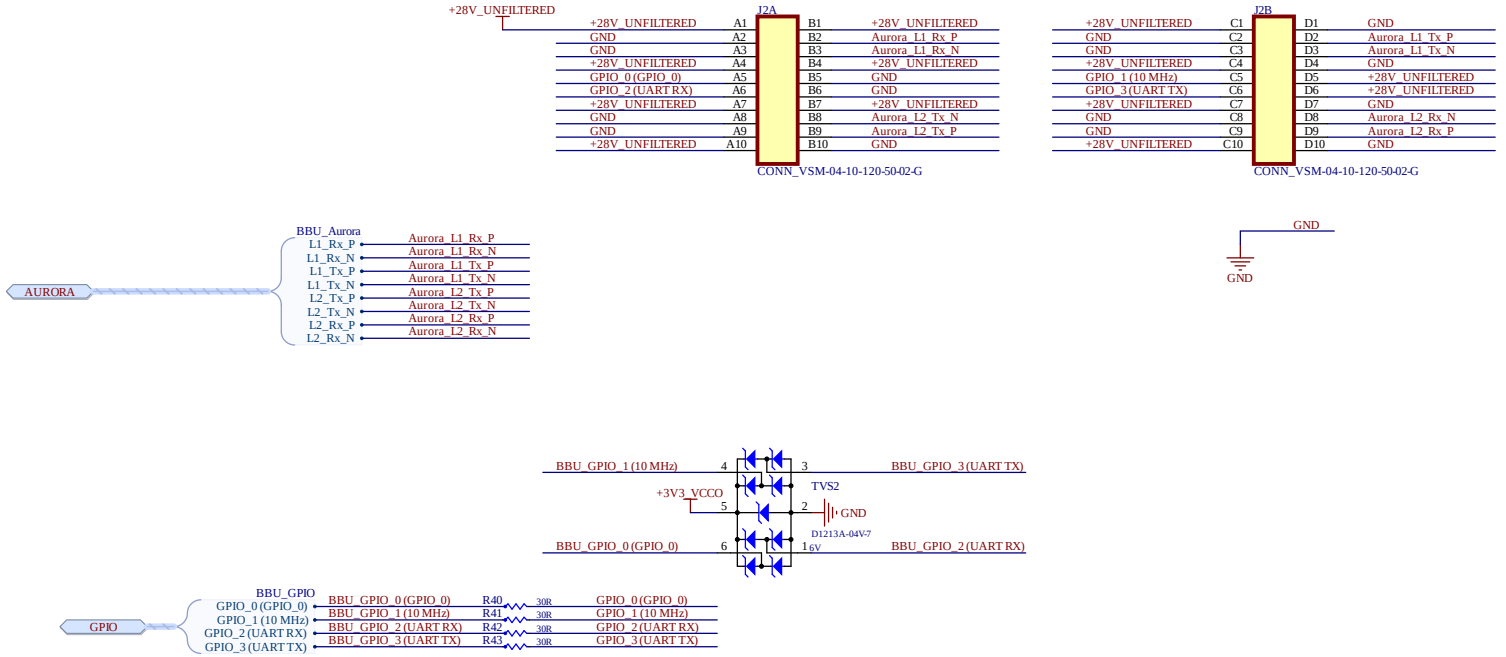
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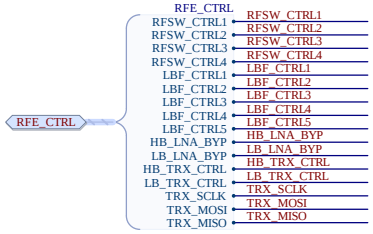
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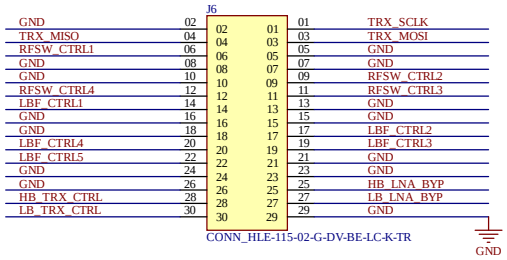


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